

# Sustainable Electrochemical Recovery of Cadmium and Nickel from Spent Ni–Cd Electrolytes and Their Interface Modification with $\text{Ca}(\text{OH})_2$

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## Abstract

This paper includes a combined electrochemical methodology of the selective recovery of cadmium and nickel in used Ni–Cd battery electrolytes and the assessment of the electrochemical activity of recovered electrodes following the  $\text{Ca}(\text{OH})_2$  surface functionalization. Cyclic voltammetry, linear sweep voltammetry, sampled current polarography and electrochemical impedance spectroscopy were used to determine the potential of deposition of Cu, Cd, and Ni in simulated solutions and real sludge extract under conditions of cyclic voltammetry, linear sweep voltammetry, and sampled current polarography. It was discovered that copper would deposit at much more negative potentials than cadmium and nickel, so it can be removed preferentially before Cd and Ni are recovered. Stainless steel was a better electrochemical behaving material and hence it was chosen as material of preference of electrode. The Cd and Ni electrodes were recovered and electrodes modified with different concentrations of  $\text{Ca}(\text{OH})_2$  (5% and 10% and 15%). Electrochemical examination gave results that 10 %  $\text{Ca}(\text{OH})_2$  modification gives optimal redox and minimum carry-over against resistance as well as enhanced durability of currents in both electrodes. The findings prove the presence of the sustainable path to metal recovery and electrode reuse, which contribute to environmentally friendly Ni–Cd battery recycling.

**Keywords:** Nickel–cadmium batteries: Electrochemical metal recovery: Cadmium and nickel electrodeposition: Calcium hydroxide surface modification: Electrochemical impedance spectroscopy.

## 1.INTRODUCTION

Nickel-cadmium (Ni–Cd) batteries have widely found application in different services such as portable electronic services, emergency lighting services, aircraft, and power backup units due to their strength as well as capability to generate high discharge rates [1,2]. But existence of cadmium in such cells causes severe environmental and health effects given that cadmium is known to be a toxic and a carcinogenic heavy metal [3,4]. Consequently, used Ni–Cd batteries present an important source of hazardous waste, unless well disposed. The fact that leakage or incorrect disposal of the electrolyte of these cells causes pollution of the land and ground water makes the recovery of cadmium a vital demand in the environment [5,6].

Electrolyte of NiCd cells is usually alkaline, which contains KOH, in which the cadmium ions dissolve when the cell is subjected to a long cycle and when it is overcharged [7,8]. The build-up of cadmium will

begin to reduce the efficiency of the battery, in addition to making it necessary to process the spent electrolytes as time goes by [9,10]. The distinction of  $\text{Cd}^{2+}$  and  $\text{Ni}^{2+}$  is however difficult because of the similarity in their ionic radii and chemical properties. In response to this, researchers have studied new extractants, combinations and combined recovery techniques that aim at improving selectivity and recoveries.

Using the waste of cadmium in the electrolytes to recycle is not only a solution of not destroying the environment but also offers economic advantages through recovery of useful item. Different hydrometallurgical processes were investigated in the recovery of cadmium, the processes used are precipitation, ion exchange and solvent extraction [11,12]. The solvent extraction is especially helpful among them, as chelating extractants, e.g., organophosphorus compounds, may employ high  $\text{Cd}^{2+}$  selectivity relative to  $\text{Ni}^{2+}$ . Hydroxide reagent or

sulfide reagent methods. Precipitation techniques have also been developed allowing cadmium salts to be further utilized [13,14]. More modern techniques combine solvent extraction with electrodeposition to allow the direct recovery of metallic cadmium out of the electrolytes.

The resulting metallic cadmium can then be used as electrodes by electrodeposition. The recovered cadmium electrodes can be characterized to make sure that they are structurally, chemically, and electrochemically suitable. Electrochemical techniques such as cyclic voltammetry and electrochemical impedance spectroscopy (EIS) give information about electrode kinetics, charge transfer resistance and stability. By taking these analyses, it is ensured that the recovered cadmium electrodes can do the same functions as commercial ones justifying the recycling process.

Recycled cadmium may be electrodeposited onto substrates in a conductive form so as to produce electrodes with a specific morphology and electrochemical properties. The long-term stability and performance of these electrodes is determined by the surface roughness, corrosion resistance, and charge transfer properties of these electrodes. Structural, electrochemical characterizations do not only verify purity, but also are used to compare the materials retrieved against commercial cadmium electrodes.

In addition to the technical aspect, the recovery of cadmium and Ni-Cd used with used Ni-Cd cells complies with the ideas of green chemistry and the circular economy. Recycling ensures that reliance on mining of virgin cadmium and nickel is minimal, there is reduction in the production of hazardous wastes and the process adheres to more strict policies on battery disposal, which are becoming more stringent each day [15,16]. The life cycle analyses also advocate sustainability of such recovery processes and prove environmental protection and economic profitability [17,18].

To conclude, the recovery of cadmium and nickel using spent Ni-Cd cell electrolytes and the subsequent nature of cadmium electrodes are a sustainable solution to the recovery of resources and management of their wastes. Closure of a metallurgical loop in the utilization of cadmium and nickel can be achieved by combining the hydrometallurgical methods with the methodology of determining the characteristics of materials. This does not only help to avoid pollution to the environment but also facilitates the reuse of cadmium in the electrode uses, thus the reuse of the cadmium resulting into ecologic and economic gains. Therefore, the current research was conducted to restore used electrolyte as well as investigate the impact of  $\text{Ca}(\text{OH})_2$  on the performance of reclaimed Cd and Ni electrodes.

## 2. EXPERIMENTAL

### 2.1 Mechanical separation

Life expired Fibre Technology Ni - Cd cells are taken to the study and this is convenient because the discussion can be done on the recovery of cadmium and nickel. All the cells are dissected separately and washed 5 times in demineralized water. The cells are broken to pieces, and the electrolyte is taken off. Hardware is sponged using DM water. This is a fluid,  $\text{Ni}^{+2}$ ,  $\text{Cd}^{+2}$  and in traces alkali. The water collected is washed and identification taken. The negative and positive electrodes are eliminated. Washing of the electrodes is done individually and the washed water established.

### 2.2 Dissolution of active material

They collect the sediments of positive and negative electrodes. These sediments are dissolved using concentrated sulphuric acid. The solution is stored independently and identified. As extractant, hydrochloric acid has been taken into consideration. 200 ml of 3.6 N hydrochloric acid per kg of sludge material was used as extractant.

### 2.3 Determination of chemical composition

Chemical composition of the spent extract determined by quantitative analysis, we have determined 4.5% of each Cd and Ni and very low percentage of Cu and silica.

### 2.4 Determination of deposition potential

The current paper is primarily concerned with the recovery of Cd and Ni only after the sludge extract since only they can be recycled in the Ni-Cd battery sector. One of the impurities that has been studied is copper removal, which has been known to corrupt electro separation of Cd and Ni. Other impurities Ca, Mg and Si do not require to be electro separated and can be disposed as hydroxides. The element Cd, Ni and Cu and this has only been calculated in the following steps against which deposition potential has been determined.

- It identifies the capacities of deposition of the technique of interest.
- Identification of deposition possibilities of the same elements in the sludge extract.

The different types of electrochemical practices used include:

- Cyclic Voltammetry (CV)
- Samples current polarography (SCP).

SCP is proven to be the appropriate method of establishing the potentials of deposition. A standard three electrodes cells set has been utilized with the size of 2 cm x 2 cm. Pt has been used as cathode. As a cathode material, platinum (Pt), stainless steel (SS), Graphite (Gr) and Titanium (Ti), in the size of the micro (diameter 1 mm) have been analysed. The

reference electrode has been taken as saturated calomel electrode (SCE). All the electro chemical experiments have been undertaken using a Zahner Zennium XC high level electrochemical work station.

## 2.5 Cell assembly for electro separation studies

The cell that will be utilized in the recovery of cadmium and nickel ions used into the simulated solutions and spent electrolyte has been fitted with the electrodes. The anode is a central Teflon cylinder of 30 mm diameter attached to the lid of Teflon on which the platinum gauze is wound around. The cell with the cathode is attached to saturated calomel electrode via a lugging that acts as reference electrode.

## 2.6 Electro separation studies with spent electrolyte

Following the establishment of optimum deposition of the copper, cadmium and nickel in the simulated solutions and spent electrolyte, the deposition of the various metals has been performed at their determined respective deposition potential in Bulk Electrolysis mode with Zahner Zennium XC electrochemical work station. Similar studies have been carried out on electro separation of the sludge extract as have been in case of stimulated solutions.

## 2.7 Deposition of Calcium hydroxide and evaluation of performance of Cd and Ni electrodes recovered from spent electrolyte

In conducting the experimental study and in determining the performance of the nickel electrode in charging, solutions of Calcium nitrate were prepared at 3 different concentrations, which were 5%, 10% and 15%. These electrodes were made by means of electrochemical impregnation process are washed in the solutions and functions of these electrodes were tested.

The additive preparation, its different concentration, processing procedures and its evaluation is discussed below.

### 2.7.1 Preparation of an active Nickelic hydroxide

It is a type of electro chemical impregnation whereby the electrode goes through electrical polarization in a solution of Nickel nitrate. Counter electrode is also inert which serves as positive electrode. In cases where the electrode over which impregnation is done is with hydroxide.

### 2.7.2 Preparation of Calcium nitrate solutions

A 5% solution of  $\text{Ca}(\text{NO}_3)_2$  is made by dissolving it in DM (Demineralized) water and then letting the mixture reach 1000 ml. Also, 10% and 15% Magnesium nitrate solutions were prepared by dissolving 100g and 150g Magnesium nitrate respectively in DM water and solution is prepared up

to 1000ml. 100 gm of KOH pellet was dissolved in the DM water (Demineralized) and solution is made to 1000ml. This is kept in a beaker of one liter capacity.

### 2.7.3 Soaking the electrodes in 5% $\text{Ca}(\text{NO}_3)_2$ solution

Six Ni electrodes, 13 cm X 7 cm were taken. A glass beaker was used to take one liter of Calcium nitrate solution. The Calcium nitrate solution was then soaked on the electrodes, and were stored in the solution of 10 minutes. The soaked electrodes were dried at ambient temperature and taken. The electrodes were dried and put once more into 10% KOH solution after 5 minutes.

The same procedure of Calcium nitrate solution retention in the electrodes was used in determining the solution prepared initially with 10% concentration level and then with 15%. The procedure was repeated and the retention of  $\text{Ca}(\text{NO}_3)_2$  in the electrodes was measured.

### 2.7.4 Conversion of Calcium nitrate to Calcium hydroxide at the surface of electrode

The electrodes were presoaked in the 10% KOH solutions (10 minutes), dried in ambient condition. Then the dried electrodes were again immersed in 10% KOH solution of 5 minutes. The wet electrodes were dried off. This is repeated with the electrodes in 10% and 15%  $\text{Ca}(\text{NO}_3)_2$  solutions respectively.

The wet electrodes were dipped in water which ran over the DM electrodes, 5 minutes. The washed electrode was now taken and washed by placing in a beaker until the washed water becomes neutral. Unless hydroxide ions are washed properly, there will be formation of carbonate and washing it off will be a tedious process. The electrodes were dried in 60°C during 1 hour after washing.

## 2.8 Electrochemical Impedance Spectroscopy

The Ni-Cd cell was made using H type glass cell which had the solution of 6M KOH. The Ni and Cd electrodes were used as the working and the counter electrode respectively. Reference electrode was Hg/HgO electrode. Cyclic voltammograms were recorded after which electrochemical impedance spectra were recorded by superimposing a sinusoidal voltage amplitude of 10 mV at an open circuit potential.

## 2.9 Cyclic Voltammetry

The cyclic voltammogram of bare and  $\text{Ca}(\text{OH})_2$  deposited nickel and cadmium electrodes were also taken to the scan rates of 20, 50 and 100 mV/S in the same cell set up used in the recording electrochemical impedance spectroscopy.

## 2.10 Chronoamperometry

Chronoamperometric experiments have been conducted at various anodic potentials on the same H type cell in order to assess the nature of the  $\text{Ca}(\text{OH})_2$  coatings on nickel and cadmium electrode.

## 3. Results and Discussion

### 3.1 Theoretical Background of Metal Deposition

The aqueous solution containing metals is electrodeposited depending on their standard reduction potentials, solution chemistry, complex formation, and electrode material. Overpotential, and complexation effects and pH of the solution make the actual deposition potential differ with respect to the standard value [19]. When the shifting reagents (complexing agents) are added, the free metal-ion concentration is changed and this causes the shift of

the deposition potential to more negative values as will be discussed in terms of Nernst equation [20]. This principle may be of special use in attaining selective electro-separation of metals that have similar reduction potentials including Cd, Ni, and Cu. Copper is also known to form selective deposition and is capable of interfering with selective recovery of Cd and Ni [21]. Therefore, its deposition characteristics were examined in more detail with the alkaline earth metals like Ca and Mg and non-metals like Si being eliminated chemically as hydroxides and were not electro-separated. Table 1 shows the deposition potentials of Ni, Cd as well as Cu in simulated solutions on platinum (Pt) electrode, stainless steel (SS) electrode, titanium (Ti) electrode and graphite (Gr) electrode under different chemical conditions.

**Table 1:** Metal deposition potential (v) for the simulated solutions for various electrodes in the presence and the absence of the shifting reagents.

| Chemical conditions of the simulated solutions | Nickel |        |        |        | Cadmium |        |        |        | Copper |        |        |        |
|------------------------------------------------|--------|--------|--------|--------|---------|--------|--------|--------|--------|--------|--------|--------|
|                                                | Pt     | SS     | Ti     | Gr     | Pt      | SS     | Ti     | Gr     | Pt     | SS     | Ti     | Gr     |
| Acidic (pH 1.2)                                | -0.95  | -0.9   | -0.8   | -0.8   | -0.7    | -0.7   | -0.8   | -0.8   | -0.15  | -0.15  | -0.225 | -0.225 |
| Citric (added)                                 | -1.0   | -0.92  | -0.8   | -0.8   | -0.74   | -0.72  | -0.72  | -0.72  | -0.16  | -0.21  | -0.21  | -0.18  |
| Tartaric (added)                               | -0.92  | -0.86  | -0.8   | -0.8   | -0.74   | -0.72  | -0.72  | -0.78  | -0.16  | -0.21  | -0.225 | -0.287 |
| EDTA (added)                                   | -1.0   | -0.92  | -0.82  | -0.82  | -0.73   | -0.75  | -0.75  | -0.759 | -0.2   | -0.26  | -0.26  | -0.287 |
| Ammoniacal (pH 10)                             | -1.097 | -1.10  | -1.08  | -1.078 | -0.73   | -0.73  | -0.73  | -0.82  | -0.22  | -0.241 | -0.241 | -0.246 |
| Ammoniacal (pH 11)                             | -1.151 | -1.138 | -1.138 | -1.138 | -0.88   | -0.874 | -0.874 | -0.901 | -0.24  | -0.261 | -0.261 | -0.246 |
| Ammoniacal (pH 11.5)                           | -1.105 | -1.102 | -1.10  | -1.108 | -0.915  | -0.915 | -0.915 | -0.907 | -0.22  | -0.217 | -0.217 | -0.229 |
| Ammoniacal (pH 12)                             | -1.197 | -1.197 | -1.197 | -1.197 | -0.907  | -0.907 | -0.907 | -0.907 | -0.29  | -0.29  | -0.29  | -0.29  |

When in an acidic environment Cd has the capability to deposit in the range of -0.7 to -0.8 V whilst Ni deposits at even lower potentials (-0.8 to -0.95 V) showing that Cd can be selectively deposited before Ni. Copper, however, displays deposition at much lower negative potentials (~0.10-0.22 V) and this proves its preferential deposition and possible interference during Cd and Ni recovery.

Pt and SS have slightly lower deposition potentials, which is due to their excellent catalytic properties and lower hydrogen overpotential amongst other electrodes. Citric acid, tartaric acid, and EDTA addition causes the depositional potentials of Cd and Ni to move to larger negative values as a result of strong complexation with the metals. The greatest negative shift is observed in EDTA, especially in the case of Ni which is corroborated by its stability constant with Ni being higher than the one exhibited with Cd. The potential of copper deposition also

changes downwards, but by far not as downwards as those of Cd and Ni; this means that it can be removed selectively before Cd and Ni is deposited. Such behaviour confirms that it is possible to use complexing agents effectively to increase the selectivity between Cu, Cd and Ni.

As the pH increases (10 to 12), there would be an increasing shift in the deposition potentials of the three metals towards the negative field. The shift in nickel is maximum since it forms stable complexes with ammonia. Cadmium exhibits moderate changes, whereas the changes in copper are not so significant. Above  $\text{pH} \geq 11$  the separation window between Cd and Ni is widened therefore stepwise electrodeposition becomes possible.

Table 2 highlights the potentials behind the deposition of the same metals in the real sludge extract, which includes a number of co-existing ions and impurities.

**Table 2:** Metal deposition potentials (v) for the sludge extract on various electrodes in the absence and the presence of the shifting reagents

| Chemical conditions of the sludge extract | Nickel |        |        |        | Cadmium |        |        |        | Copper |        |        |        |
|-------------------------------------------|--------|--------|--------|--------|---------|--------|--------|--------|--------|--------|--------|--------|
|                                           | Pt     | SS     | Ti     | Gr     | Pt      | SS     | Ti     | Gr     | Pt     | SS     | Ti     | Gr     |
| Acidic (pH 1.2)                           | -1.002 | -1.105 | -1.09  | -0.986 | -0.755  | -0.802 | -0.761 | -0.586 | -0.477 | -0.474 | -0.474 | -0.360 |
| Citric (added)                            | -1.148 | -1.197 | -1.115 | -1.135 | -0.807  | -0.824 | -0.825 | -0.838 | -0.505 | -0.507 | -0.520 | -0.495 |
| Tartaric (added)                          | -1.153 | -1.197 | -1.119 | -1.153 | -0.81   | -0.825 | -0.825 | -0.817 | -0.515 | -0.525 | -0.487 | -0.495 |
| EDTA (added)                              | -1.156 | -1.197 | -1.135 | -1.135 | -0.733  | -0.808 | -0.816 | -0.816 | -0.482 | -0.482 | -0.500 | -0.482 |
| Ammoniacal (pH 10)                        | -1.16  | -1.197 | -1.117 | -1.119 | -0.81   | -0.877 | -0.871 | -0.816 | -0.248 | -0.253 | -0.248 | -0.243 |
| Ammoniacal (pH 11)                        | -1.197 | -1.197 | -1.197 | -1.197 | -0.879  | -0.871 | -0.871 | -0.816 | -0.512 | -0.512 | -0.461 | -0.484 |
| Ammoniacal (pH 11.5)                      | -1.174 | -1.197 | -1.174 | -1.197 | -0.935  | -0.928 | -0.928 | -0.928 | -0.490 | -0.494 | -0.484 | -0.443 |
| Ammoniacal (pH 12)                        | -1.174 | -1.197 | -1.174 | -1.197 | -0.935  | -0.928 | -0.928 | -0.928 | -0.494 | -0.494 | -0.484 | -0.443 |

Deposition potentials were found to be a little negative in the case of acidic sludge extract (pH 1.2), then in simulated solutions. It is explained by the fact that at high negative potentials, copper still deposits relative to Cd and Ni, which is due to matrix effects [ionic strength, competing ions, partial complexation of residual organic matter] [22]. This confirms that it will be feasible to remove copper ahead of Cd and Ni. The tendencies of simulated solutions are stored in the sludge extract, but the changes are more enhanced. Again, EDTA induces maximum negative shift particularly in Ni. This attests to the strength of the electrochemical separation plan even with complicated matrices. Citric and tartaric acids are moderate changes and might be utilized where there is need to adhere to an environmental compatibility. Ni has the lowest deposition potential under ammoniacal conditions (pH 10-12) with a deposition potential of ( $\sim -1.19$  V), Cd deposition potential ( $\sim -0.90$  to  $-0.93$  V) then follows, and Cu has a deposition potential many less negative ( $\sim -0.24$  to  $-0.49$  V). Such a broad range of deposition potentials offers a great operating range of selective recovery. In both tables Pt and SS electrodes also depict a low deposition potential that is always slightly negative than that of Ti and graphite, which point to faster kinetics in electron transfer. Although graphite is inert, it possesses higher overpotentials. SS can provide a good balance between cost and performance, and thus it can be used in large-scale applications.

The ability to deposit copper (Cu) at much lower negative potentials than Cd or Ni in both simulated solutions and actual sludge extract are systematic assessments of the deposition potentials of these three metals in both solutions, and should not be mistaken for the presence of solution effects that can significantly influence copper (Cu) deposition rates in solution. This facilitates its favorable separation thus eradicating its established interference with electro-separation of Cd and Ni.

Introduction of shifting reagents like citric acid, tartaric acid and EDTA was successful in modifying the deposition behaviour of the target metals because the changing reagents complexed with the target deposition potentials and shifted their values towards

a more negative position. Among them, EDTA yielded the largest negative shift, especially in Ni which leads to an increase in separation between Cd and Ni. Selectivity was further enhanced through the use of ammoniacal media (pH 10-12) that stabilized metal-ammine complexes with Ni having the lowest potential deposition potential, Cd being second and Cu electrochemically distinct.

Simulated solutions and sludge extract analysis showed comparable analysis trends in terms of deposition although more overpotential was needed in the sludge matrix with competing ions and matrix effects. This establishes the fact that the optimized electrochemical conditions can be really applied in the intricate natural systems. Platinum and stainless steel were the two best electrode materials that were investigated because they displayed good deposition properties with reduced overpotentials and stainless steel has a benefit of being practical to use in large-scale applications.

On the whole, the research creates a valid electrochemical method of selective removal of Cu and subsequent recovery of Cd and Ni making them potentially re-utilizable in Ni-Cd battery sector. Other impurities like Ca, Mg and Si, which are non-interfering, can be safely disposed as hydroxides and they do not require any electro-separation. The results are a good scientific foundation in the creation of both sustainable and resource efficient metal recovery process of the industrial sludge.

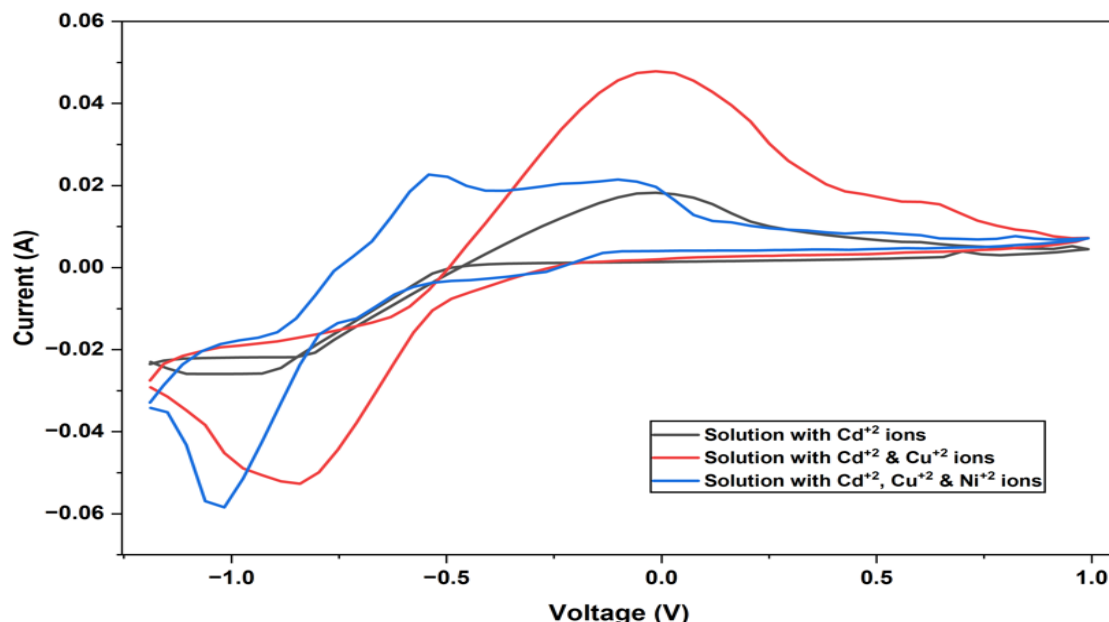
Cyclic voltammetry is an effective electroanalytical method in studying the redox characteristics, reversibility, and kinetics of deposition of metal ions on the surface of various electrodes. The potential of the working electrode of CV is swept linearly with time in the forward (cathodic) and reverse (anodic) direction, and the current response received has information on the reduction, nucleation, growth, and stripping processes of the metal ions.

Cyclic voltammogram obtained with titanium (Ti) electrode of the simulated solution has clearly defined cathodic peaks that indicate the reduction of metal ions. The cathodic current commences at considerably lower potentials depicting the increased overpotential price to deposit on Ti. Such a

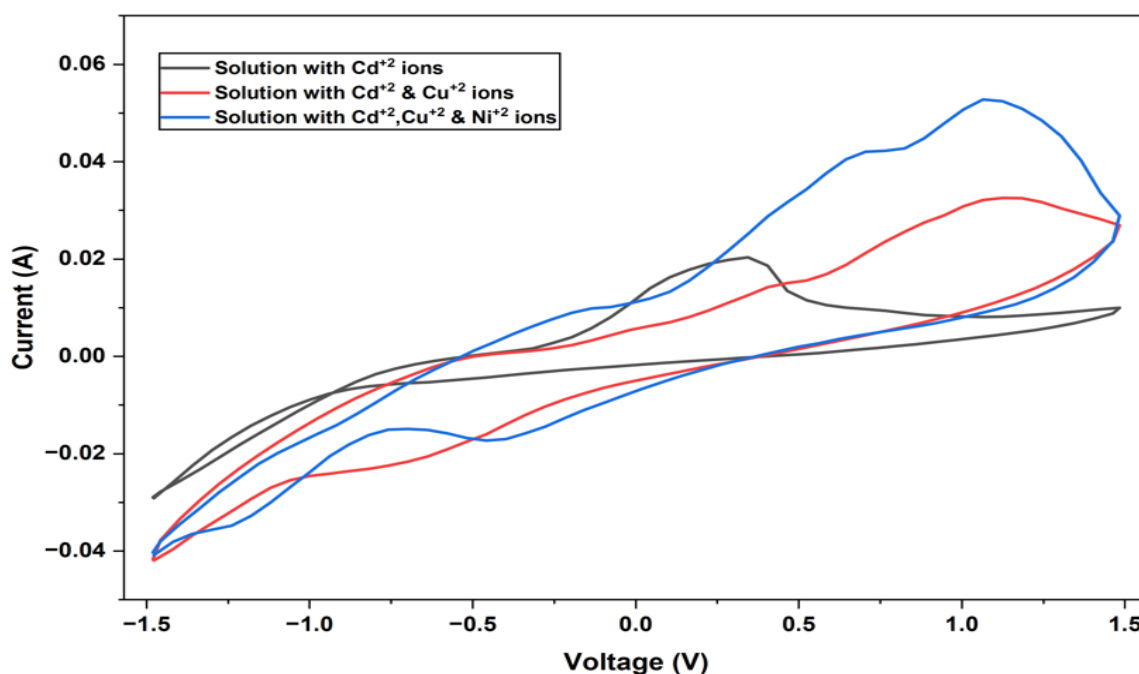
phenomenon is explained by the existence of a passive  $\text{TiO}_2$  layer that lowers the kinetics of electron transfer.

Cd cathodic peak is witnessed before the Ni one, which affirms the thermodynamic preference of Cd deposition. Closing is possible at significantly smaller

negative potentials with copper reduction very clearly seen to be separate at Cd and Ni. On the reverse scan, anodic peaks of stripping of metals are seen, albeit with broader peaks, indicating a behavior that is quasi-irreversible.



**Figure 1.** Cyclic voltammogram of the simulated metal ion solution recorded using a titanium (Ti) working electrode, illustrating the cathodic deposition and anodic stripping behavior of Cu, Cd, and Ni under the applied potential scan.



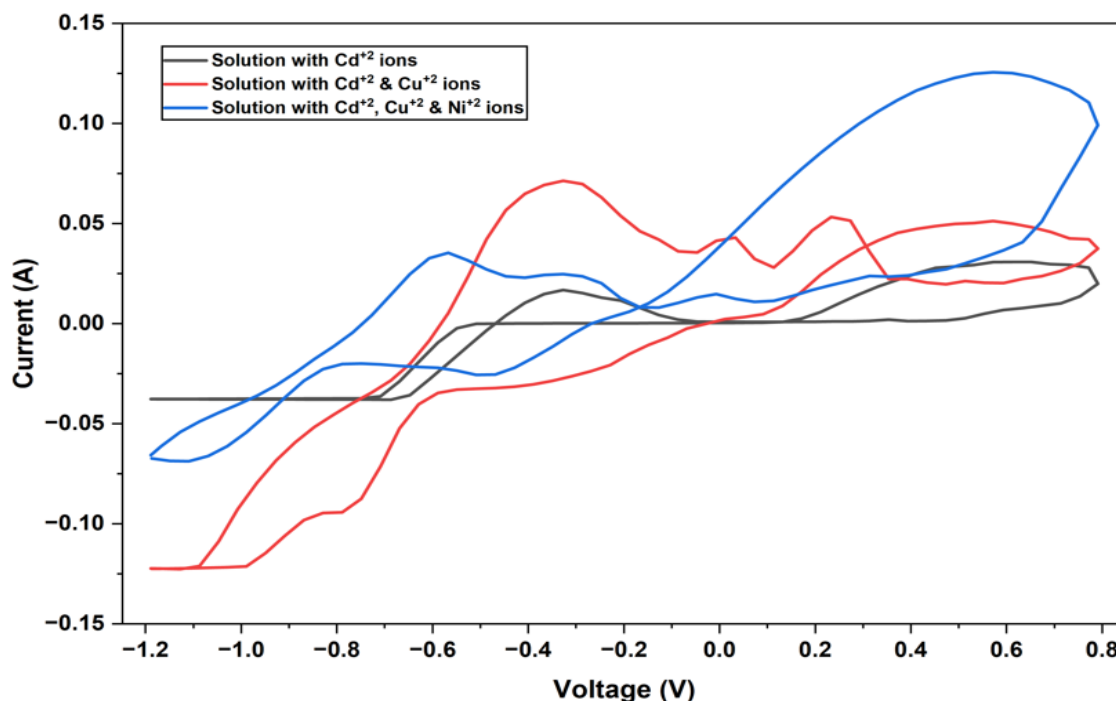
**Figure 2.** Cyclic voltammogram of the simulated metal ion solution obtained with a graphite (Gr) working electrode, showing diffusion-controlled reduction and oxidation peaks of Cu, Cd, and Ni.

The cyclic voltammogram taken through graphite (Gr) electrode has broader and less sharp redox peaks than Ti and SS electrodes. This is mainly attributed to porous nature and the rough surface of graphite that

favours the processes that rely on diffusion. At positions that are more negative, we find cathodic peaks of Cd and Ni reduction, which represents an increased resistance to nucleation. The Copper

reduction is observed at lower negative leading to a comparatively higher current response which proves that it is preferentially deposited. The anodic stripping peaks are not so intense and have a wide separation of the peaks with the cathodic ones, there

the reversibility and surface adsorption effects are not good. Although these limitations exist, graphite has chemical inertness and corrosion resistance properties.



**Figure 3.** Cyclic voltammogram of the simulated metal ion solution recorded using a stainless steel (SS) working electrode, demonstrating well-defined reduction and stripping peaks suitable for selective electro-separation of Cu, Cd, and Ni.

The stainless steel (SS) electrode is the one with the most clearly defined and intense cathodic and anodic peaks of all the three electrodes investigated. Reduction of the metal ions starts at the less negative potentials than Ti and graphite and it proves that there is an increase in the rate of electron-transfer between the electrode and the current carrier. The cathodic peak of copper appears at much less negative potentials, Cd, and Ni, at increasingly more negative potentials. The possibility of step-wise electro-separation is confirmed by this clear peak separation. Anodic stripping peaks are stiffer and nearer to the cathodic peaks which indicate that the electrochemical reversibility is higher.

The Cyclic voltammetric studies reveal that:

- One, copper deposits at by far less negative potentials than Cd and Ni on all electrodes.
  - Cd deposition comes before Ni deposition thus allowing selective recovery.
  - The working electrode that is ideal in a large-scale electro-separation in industry is stainless steel.
- The data of the results presented in the CV is very indicative of deposition potential data also obtained previously and support the electrochemical approach used on selective recovery of Cd and Ni with sludge-derived systems.

Comparative evaluation of the responses during the CV shows that the electrochemical performance trend is as follows:

Stainless steel > Titanium > Graphite

The Ti electrode exhibits, distinct separation between practices of reduction of Cu, Cd and Ni, greater cathodic overpotentials and is selective in deposition under regulated potential situations. Diffusion-controlled metal reduction in the graphite electrode is observed to be of better quality in terms of the peak separation ( $\Delta E_p$ ), reduced reproducible quality and reduced reproducible stability than in the case of metallic electrodes, whereas reduction metal ion on stainless steel demonstrates lower hydrogen overpotential, favoured surface roughness and excellent metal-substrate contact strength.

Stainless steel provides the most appropriate compromise of catalytic efficiency, maximum resolution and practicality. Titanium although chemically stable needs bigger overpotentials but graphite is worse at poor peak definition and irreversibility.

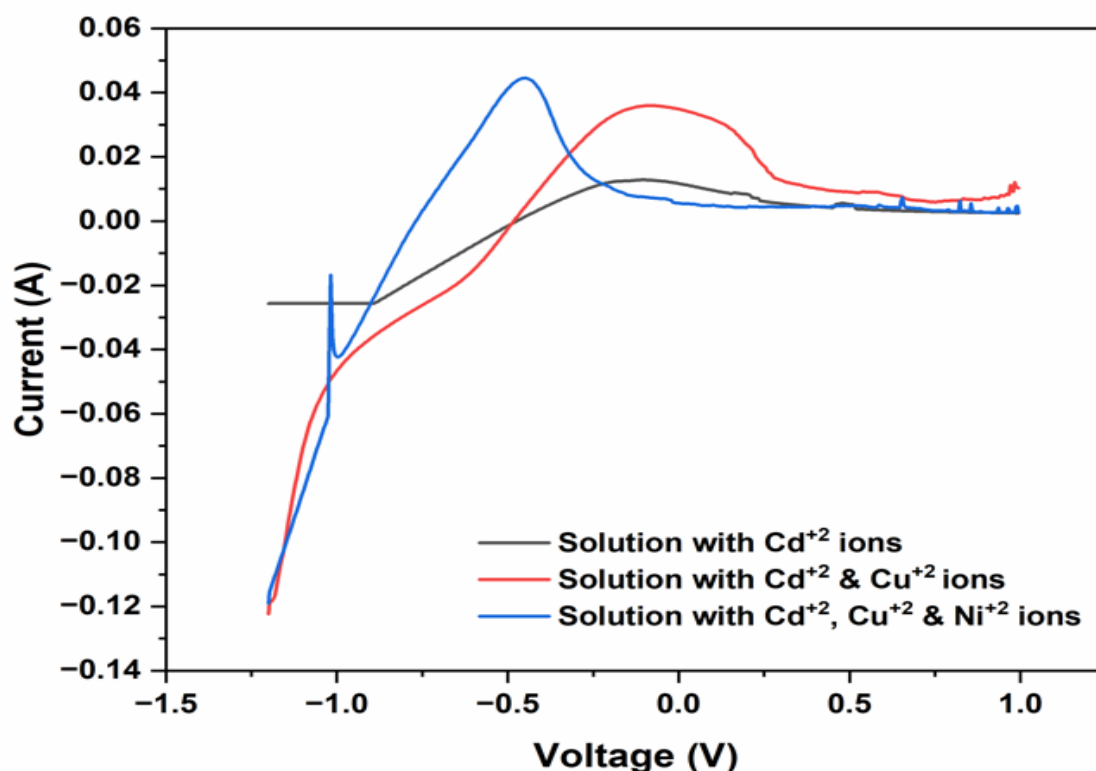
Linear sweep voltammetry (LSV) is a unidirectional electrochemical method where the electrode potential is purposefully varied with time in a cathodic direction, and the resultant response in



terms of current. LSV has found specific application in depositing potentials as suspected by CV and determining the practicability of electrode materials in unrestricted metal deposition.

The LSV observed with titanium (Ti) electrode portrays that the cathodic current increases gradually with potential specified implicating delayed formation of metal ion reduction. Start-up of cathodic current at relatively lower potentials is found to establish the increased overpotential requirement

in metal deposition on Ti. Such behaviour is strongly consistent with the CV results in which, cathodic peaks of Cu, Cd, and Ni had been conditioned to more negative potentials and lower currents peaks were noted in result of surface passivation. The slow increase in the current at LSV also confirms that the passive  $\text{TiO}_2$  layer is a limiting factor of the electron-transfer kinetics and titanium is not specifically favourable to facilitate efficient metal deposition in spite of its chemical **stability**.



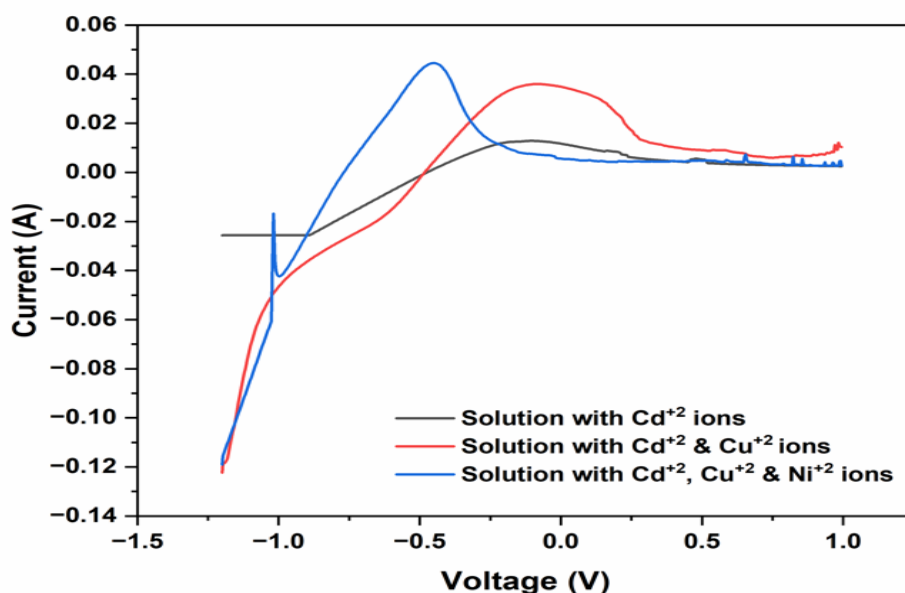
**Figure 4.** Linear sweep voltammogram of the simulated metal ion solution recorded using a titanium (Ti) working electrode at a scan rate of  $20 \text{ mV s}^{-1}$ , highlighting the higher overpotential required for metal ion reduction.

In the LSV curve, there are no sharp inflections, and the cathodic current of the graphite electrode increases in a smooth but broad manner. The onset potentials involving metal deposition are displaced further in negative direction than stainless steel and titanium. This observation is highly associated with the behaviour of CV where:

- Wide and unresolved cathodic peaks were seen.
- Greater separation of the peaks ( $\Delta E_p$ ) was an indication of irrevocable behavior.

The slow increase in the current gradient over time is an indication of a diffusion-dash deposition, and the nucleation is not very efficient. Even though graphite cannot be chemically reacted, and has corrosive properties, the electrochemical response attests that it is not as active in selective and controlled deposition of Cd and Ni.





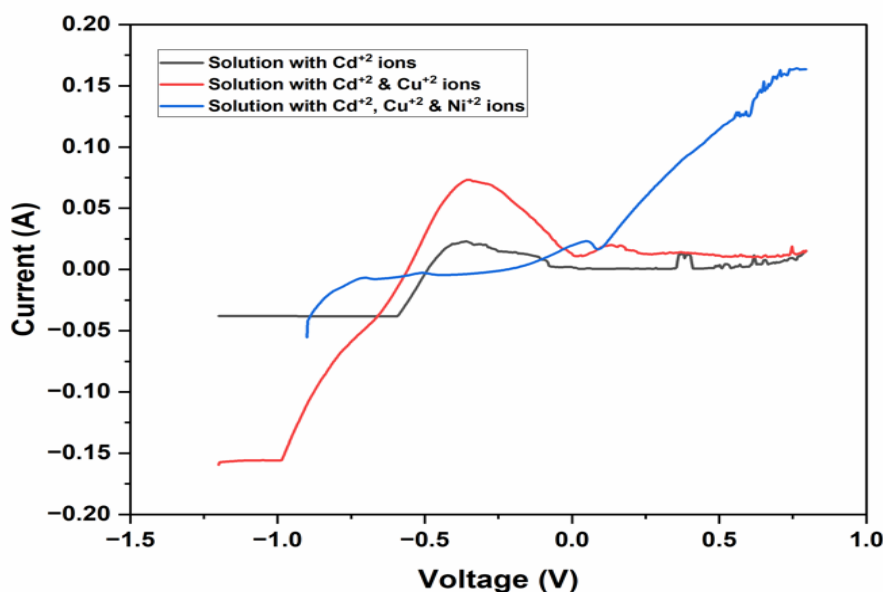
**Figure 5.** Linear sweep voltammogram of the simulated metal ion solution recorded using a graphite (Gr) working electrode at a scan rate of  $20 \text{ mV s}^{-1}$ , showing the cathodic current response associated with metal ion deposition.

The electrode is a stainless steel (SS) structure of which the cathodic current rises significantly with relatively low negative potentials pointed out in the LSV curve. Current onset will take place sooner than in Ti and graphite, which means that metal ion reduction requires less energy.

This behaviour is directly indicated in observations of the CV where:

- Cathodic peak potentials were least in the negative.

- Peak currents were highest
  - Redox peaks were sharp and pseudo reversible.
- The observation of high current rate with the deposit of metals is the evidence of the quick nucleation and growth of deposited metals and justifies that the stainless steel offers the best surface on which metals can be deposited among the electrodes examined [23].



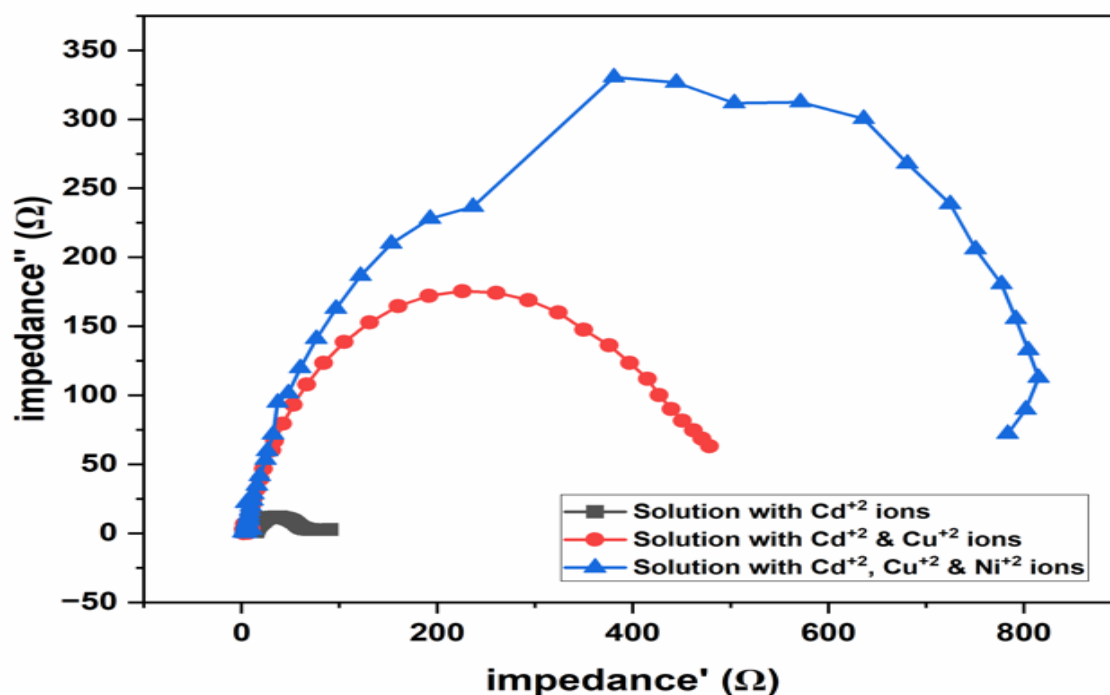
**Figure 6.** Linear sweep voltammetric response of the simulated Cu–Cd–Ni solution on a stainless-steel electrode at  $20 \text{ mV s}^{-1}$ , showing early onset potential and enhanced cathodic current suitable for efficient electrodeposition.

One of such techniques is the electrochemical impedance spectroscopy, which is employed in the study of charge-transfer resistance, mass-transfer

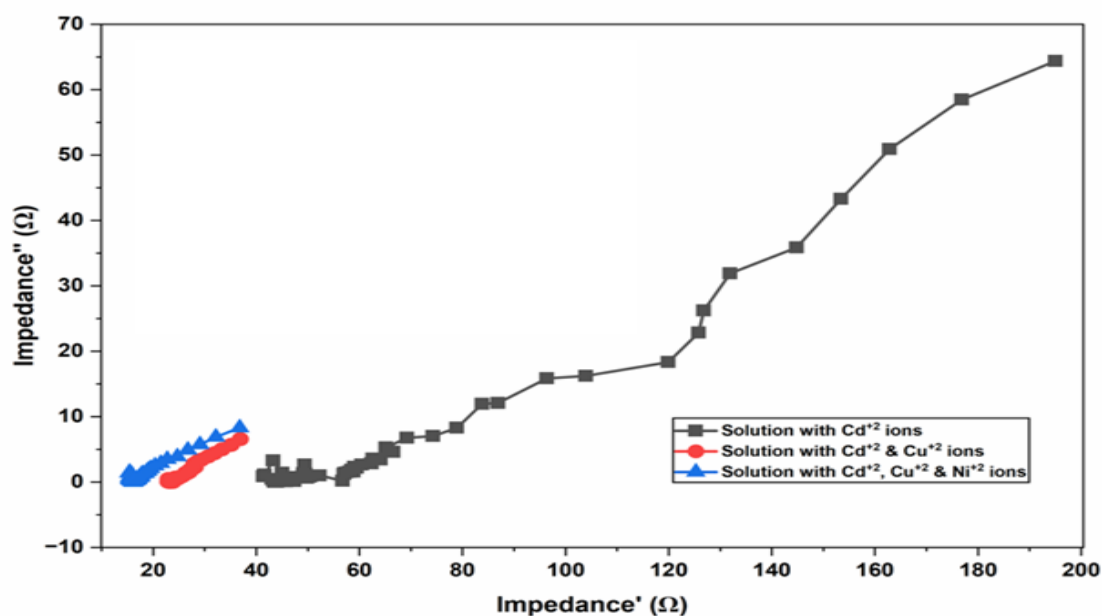
processes, and double-layer capacitance during electrode at electrolyte interface [24]. The impedance data are often plotted in Nyquist plots where the real

portion ( $Z'$ ) and the imaginary portion ( $Z''$ ) indicate the resistive and capacitive aspects respectively [25].

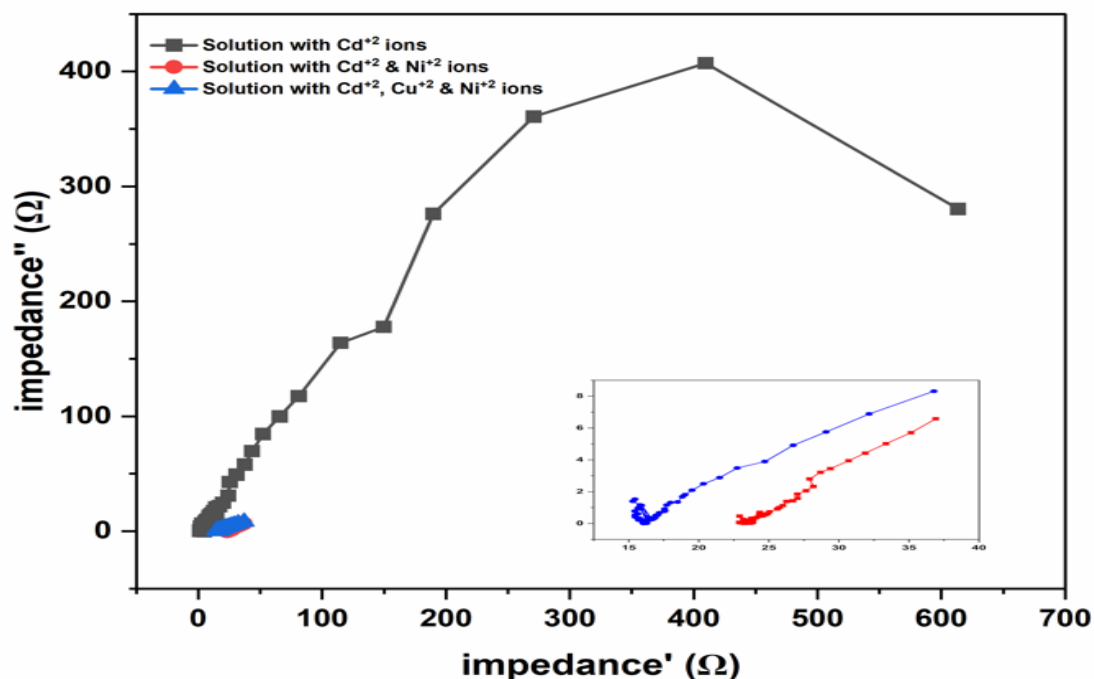
The Ti, graphite and Stainless steel electrode Nyquist plots are shown in the Figure.7,8 and 9 respectively.



**Figure 7.** Nyquist plot obtained for the simulated metal ion solution using a titanium (Ti) working electrode, indicating intermediate charge-transfer resistance influenced by surface passivation.



**Figure 8.** Nyquist plot obtained for the simulated metal ion solution using a graphite (Gr) working electrode, showing high charge-transfer resistance and diffusion-controlled impedance behavior.



**Figure 9.** Nyquist plot obtained for the simulated metal ion solution using a stainless steel (SS) working electrode, illustrating low charge-transfer resistance and enhanced electron-transfer kinetics suitable for metal deposition.

The titanium electrode has a semicircle of moderate size in its Nyquist plot which is not as large as the one of graphite but much larger than that of stainless steel. This is a pointer of a middle charge-transfer resistance [26].

The impedance response is very much affected due to the availability of a passive  $\text{TiO}_2$  layer has a barrier effect to the transfer of electrons [27]. Despite the fact that titanium can be used with a high overpotential because it has a great deal of corrosion resistance, this passive coating enhances interfacial resistance and thus, higher overpotentials are necessary to deposit the metal. This is observed at lower frequencies, in terms of a weak diffusion tail which shows that both the charge-transfer process and diffusion process are seen to contribute to the total impedance.

The Nyquist graph of the graphite electrode has a huge semicircle in the high- to medium-frequency range, which is then followed by a slanting linear section in the lower frequency range. The number of semicircles is large, which denotes that charge-transfer resistance is high, hence the slow rate of electron-transfer at the graphite-solution interface.

The low-frequency linear tail is an indication that diffusion concepts are predominant in the overall deviance of impedance [28]. The porous and heterogeneous surface of graphite makes the situation unequal distribution of current and low nucleation efficiency. This makes graphite lack the electrochemical reversibility as well as suitability in regulated metal electrodeposition.

However, the stainless steel electrode Nyquist plot shows a small and sharp semicircle, which indicates that the electrode has a small charge-transfer resistance and fast electron-transfer reactions. The smaller semicircle diameter is a sign that the reduction of the metal ions can easily take place on the stainless steel surface. The low-frequency part is associated with a low diffusion impedance implying that the deposition process is mostly controllable by charge-transfer than it is limited by diffusion [29]. This action is a direct indication of the experimental results of CV and LSV. The lower hydrogen overpotential, favourable surface roughness of stainless steel and its stable conductive oxide layer are among the factors that encourage good nucleation and growth of the metals.

According to the comparative electrochemical analysis, stainless steel was characterized as the most convenient working electrode in deposition of the metal ions of preference as opposed to titanium and graphite. Analysis of cyclic voltammetry has shown that stainless steel has lower cathodic deposition capability, higher peak current and sharper and better-defined redox currents of Cu, Cd and Ni, which implies that stainless steel has a superior electron-transfer kinetics and nucleation characteristics. Conversely, because of surface passivation with the oxide layer Titanium needed higher overpotentials to pass current and Graphite reached wider peaks, greater separation of the peaks and reductions of current response, typical of diffusion-controlled and processes that are not

reversible. It has also been demonstrated that the reduced maximum potential separation and enhanced reversibility on stainless steel further attests to its enhanced performance in electrochemical applications. Besides its electrochemical benefits, stainless steel has mechanical strength, resistant to corrosion, economical, and scalable attributes; it is more fitting in practical and industrial scale electro-deposition of metal ions through complex solutions. Therefore, a stainless steel was chosen as an electrode material of choice in the deposition and recovery of metal ions in the future.

All in all, the three electrochemical methods conclude that stainless steel provides the most preferable electrochemical environment to be used when doing metal ion deposition, this includes low overpotential, high current efficiency, quick charge transfer, and good electrochemical reversibility. Titanium, in spite of its stability in chemistry, is affected by the

passivation-based limitations whereas graphite is limited by its high resistance and diffusion-controlled characteristics. The fabulous concordance between CV, LSV, and EIS allows a good electrochemical foundation to change stainless steel as the electrode material of choice to achieve an efficient and selective electrodeposition of metal ions in the current system.

### 3.2 Surface modification of electrode fabricated from spent electrolyte and electrochemical evaluation of its performance

Modification of metal electrodes by alkaline earth hydroxides e.g. calcium hydroxide [ $\text{Ca}(\text{OH})_2$ ] changes the electrochemical interface through the formation of a thin, partially isolating though ion permeable surface. This layer is capable of affecting the distribution of surface charges as well as, double-layer capacitance, charge-transfer resistance, and nucleation and growth kinetics of metal deposition.

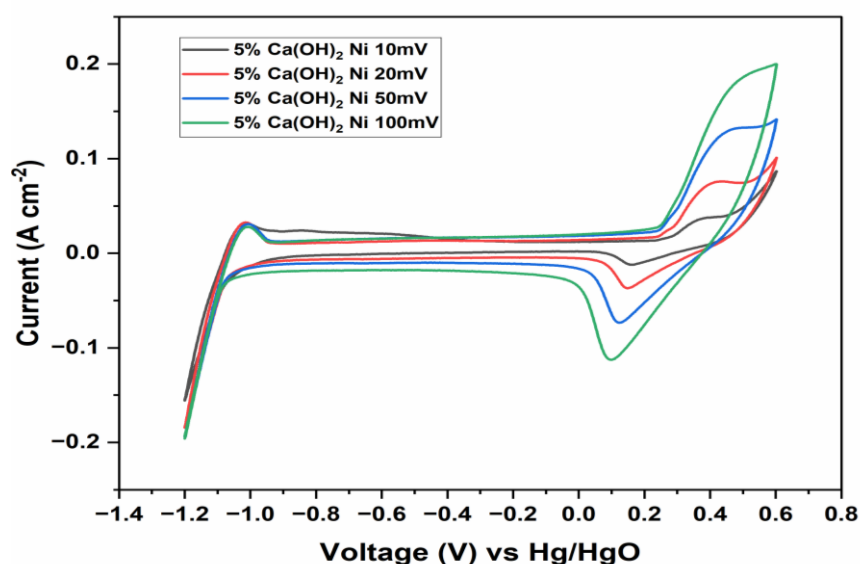


Figure 10. Cyclic voltammetric response of Ni electrode modified with 5%  $\text{Ca}(\text{OH})_2$ .

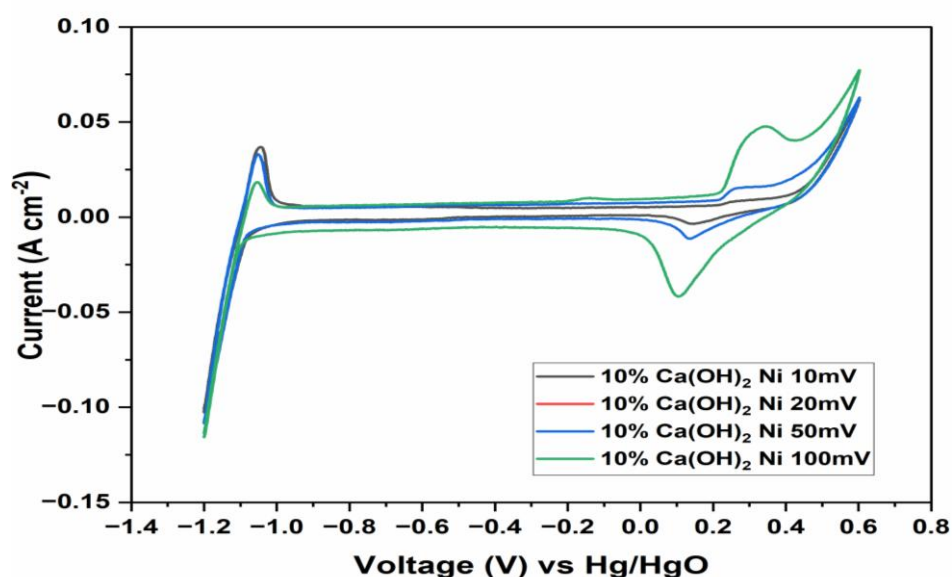


Figure 11. Cyclic voltammetric response of Ni electrode modified with 10%  $\text{Ca}(\text{OH})_2$ .

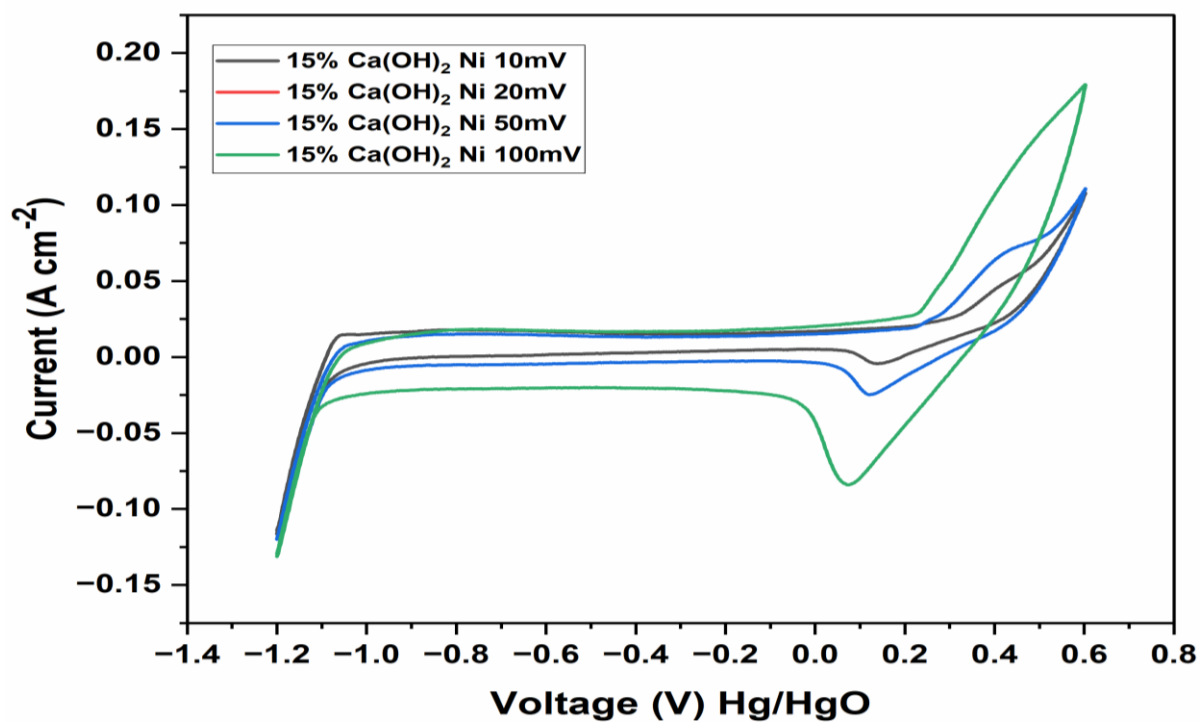


Figure 12. Cyclic voltammetric response of Ni electrode modified with 15%  $\text{Ca(OH)}_2$ .

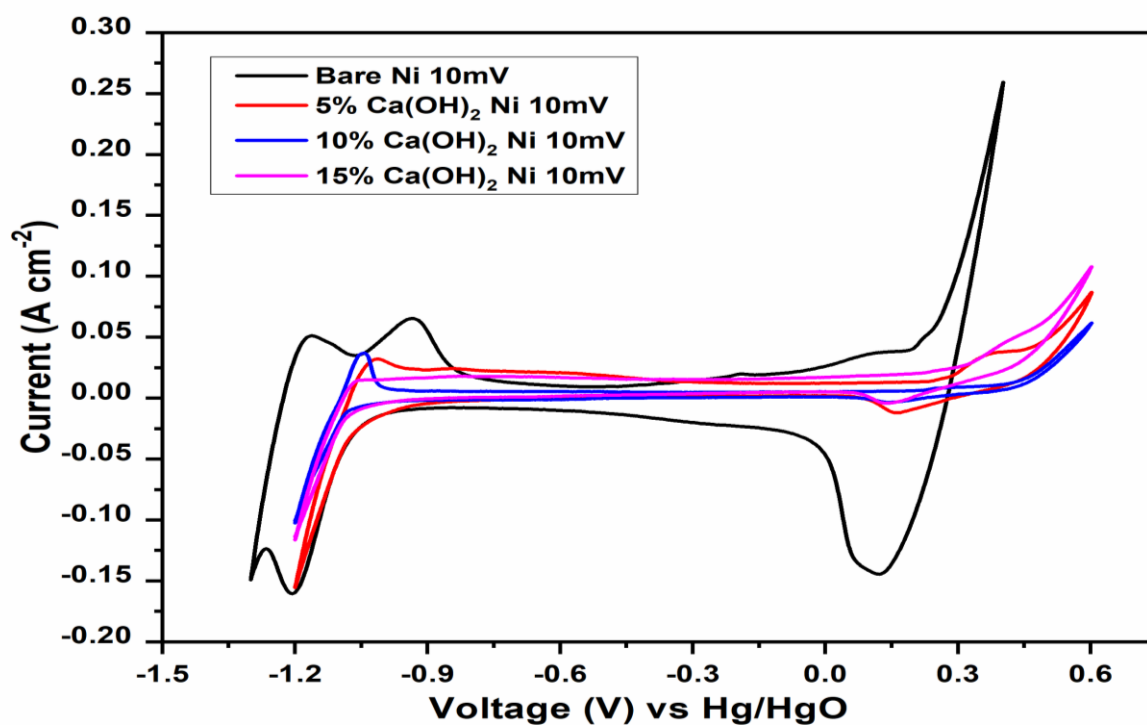


Figure 13. Cyclic voltammetric response of Ni electrode modified with 5%, 10% & 15%  $\text{Ca(OH)}_2$  at 10 mV/S scan rate.

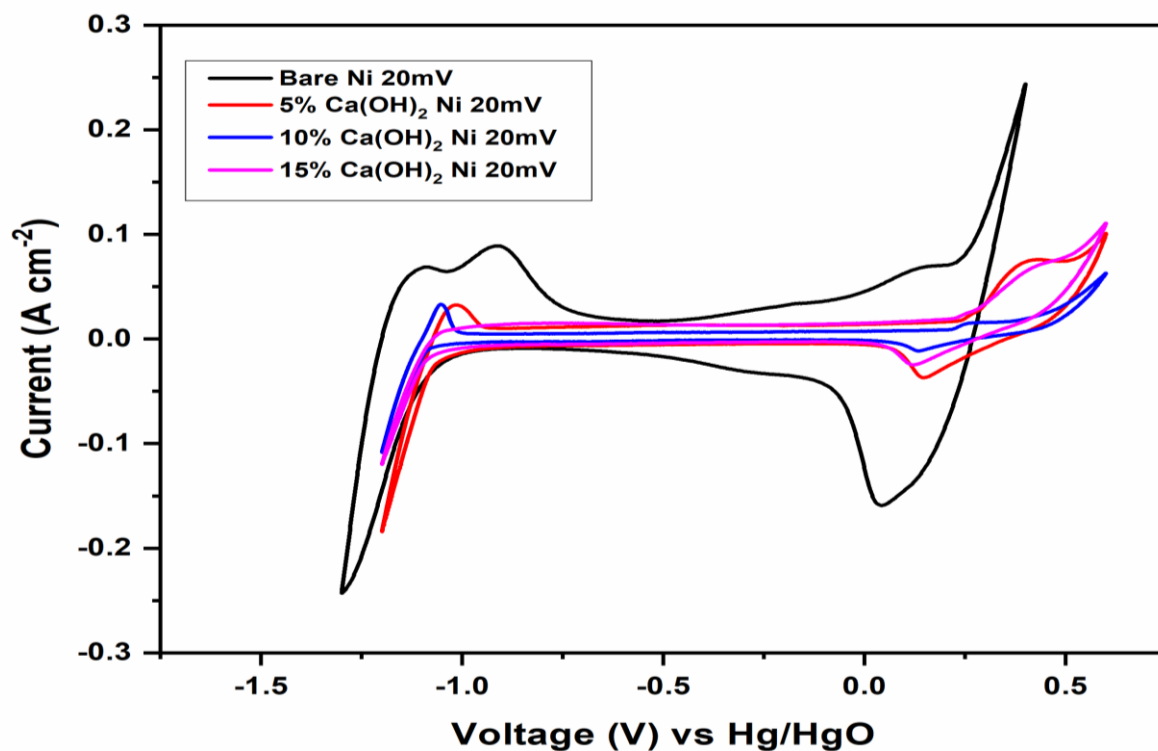


Figure 14. Cyclic voltammetric response of Ni electrode modified with 5%, 10% & 15%  $\text{Ca(OH)}_2$  at 20mV/S scan rate.

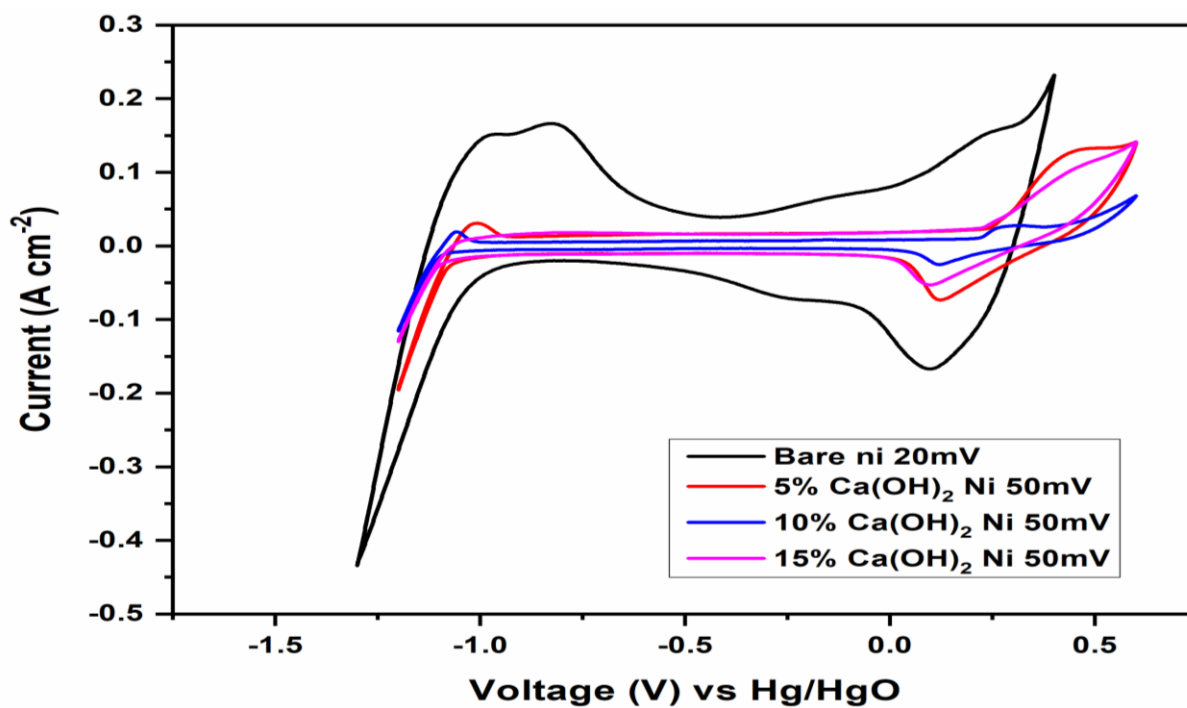
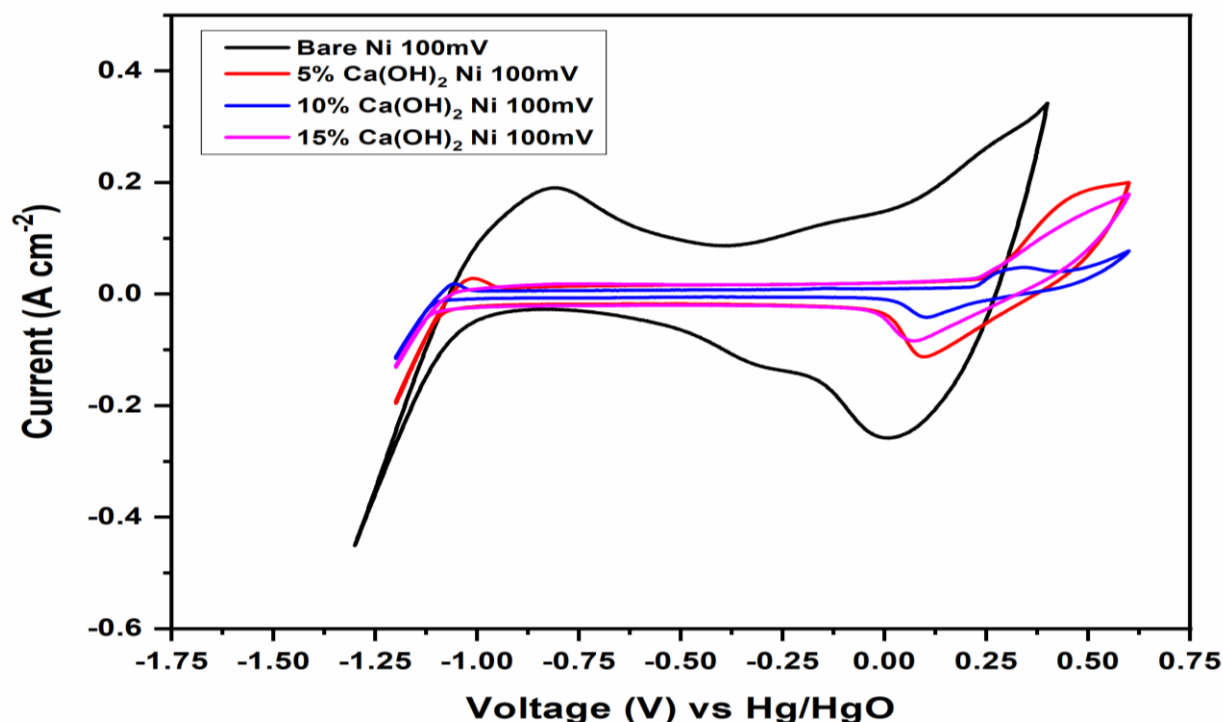


Figure 15. Cyclic voltammetry response of Ni electrode modified with 5%, 10% & 15%  $\text{Ca(OH)}_2$  at 50mV/S scan rate.



**Figure 16.** Cyclic voltammetric response of Ni electrode modified with 5%, 10% & 15%  $\text{Ca(OH)}_2$  at 100mV/S scan rate.

The cyclic voltammetry (CV) responses of nickel on stainless steel and surface-modified with  $\text{Ca(OH)}_2$  (Figures 10-16) is a clear indication of the redox manner of nickel electrodes in alkaline electrolytes. The anodic and cathodic peaks observed are related to the reversible nickel hydroxide to nickel oxyhydroxide transformation which forms the basis of charge storage of the Ni-Cd batteries. The redox reaction may be represented as follows [30]



The intensity, position and existence of these redox peaks give some idea about the reverse of electrochemical activity, reversibility and the efficiency of the modified electrodes [31]. The CV response of the nickel electrode with 5%  $\text{Ca(OH)}_2$  added in it is presented in figure 10. Clear anodic and cathodic peaks are formed, which serve as evidence of the electrochemical accessibility of the  $\text{Ni(OH)}_2/\text{NiOOH}$  couple. Yet, the comparative use of lower peak current and a bigger peak to peak separation ( $\Delta E_p$ ) indicates moderate charge-transfer resistance and partial use of active nickel sites. As the content of  $\text{Ca(OH)}_2$  rises to 10% (Figure 11), a great improvement in the density of the peak currents and a decrease in the separation of the peaks are observed. It is a sign of high electrochemical reversibility and the acceleration of electro-transfer. The optimized layer of  $\text{Ca(OH)}_2$  is likely to improve the availability of  $\text{OH}^-$  ion at the electrode-electrolyte interface, wettability of the surface, and uniform deposition of the electroactive nickel hydroxide species.

Conversely, when the proportion of the  $\text{Ca(OH)}_2$  content is further increased to 15% (Figure 12), the peak current is reduced, and the redox peaks become slightly broader. This degradation can simply be due to the high deposition of  $\text{Ca(OH)}_2$  which might partially occlude active nickel sites and provide an insulating effect thus hindering effective transfer of electrons.

These results suggest that 10%  $\text{Ca(OH)}_2$  alteration offers the best balance of ionic conductivity and electronic transportation leading to the best electrochemical execution.

The scan-rate-dependent CV profiles (Figures 13-16) also give additional interpretation of the reaction kinetics and charge storage mechanism. Figure 13 was constructed at a low rate of scanning ( $10 \text{ mV s}^{-1}$ ) and therefore, all the electrodes exhibit sharp and symmetric redox peaks indicating close-equilibrium conditions and active diffusion of  $\text{OH}^-$  ions in the electrode structure. The largest enclosed CV area and hence the highest charge storage capacity is the one with 10%  $\text{Ca(OH)}_2$  modified electrode.

There are builds up as the scan rate is highed to 20 and  $50 \text{ mV s}^{-1}$  (Figure 14 and 15) to result in corresponding anodic and cathodic peak currents accompanied by small shifts in the peaks to higher potential and lower potential respectively. This response is an indication of a more polarized and kinetically restricted response at fast sweep rates. However, the electrode with 10% of  $\text{Ca(OH)}_2$  comparison with the rest of the compositions has



more peak currents, which confirm its increased rate capability.

Peak broadening and enhancement in peak separation are more noticeable at a high scan rate of  $100 \text{ mV s}^{-1}$  (Figure 16), especially with 5% and 15%  $\text{Ca(OH)}_2$  electrodes. Diffusion bottlenecks and reduced ruling kinetics of polarization are signified by these effects. It is important to highlight that, even after the electrode containing 10% of  $\text{Ca(OH)}_2$  is modified, the transfer of redox peaks is observable, which demonstrates the high stability of its structure and speed of electrochemical reaction.

The relationship between the peak current ( $i_p$ ) and scan rate ( $v$ ) indicates a composite control mechanism between surface-confined redox reactions and diffusion, which is usually modeled as [32]

$$i_p \propto v^{1/2} \quad (2)$$

This is characteristic of alkaline systems using battery type nickel electrodes.

It is possible to attribute the following synergistic effects to the improved electrochemical performance of the  $\text{Ca(OH)}_2$  modified electrodes. The  $\text{Ca(OH)}_2$  raises the local alkalinity of the electrode surface, stabilizes the  $\text{Ni(OH)}_2/\text{NiOOH}$  redox couple, and  $\text{OH}^-$  ion movement. Besides, the modification of the surface enhances electrolyte diffusion and represses platina passivation throughout the repeated cycles. But overloading  $\text{Ca(OH)}_2$  will cause lowering of the electronic conductivity and inability to access electroactive nickel sites, as was found in case of 15%  $\text{Ca(OH)}_2$  -modified electrode. This lays weight to optimization of levels of surface modification in order to attain optimal electrochemical efficiency.

The CV data show that the surface modification of  $\text{Ca(OH)}_2$  has a strong effect on the electrochemical kinetics and charge storage characteristic of nickel electrodes. The highest redox performances, the enhanced reversibility of the macro-structure, and the superior rate performance of the studied compositions (the nickel electrodes modified by 10% of  $\text{Ca(OH)}_2$  ions) may offer the highest potential of the studied material in the high-performance Ni-Cd battery applications. The improved electrochemical properties exhibited in CV are anticipated to be reflected by the increased efficiency of the charge-discharge under working conditions.

Electrochemical impedance spectroscopy has been used to clarify the interface characteristics of the charge-transfer and mass-transport of  $\text{Ca(OH)}_2$ -modified nickel electrodes (Figure 17). The general shape of all Nyquist plots is the presence of a high-frequency intercept on the real-axis, a depressed semicircle in the mid-frequency region and an inclined line at low frequencies. This phenomenon is typical of nickel electrode working in alkaline electrolyte and reflects the joint contribution of solution ohmic resistance ( $R_s$ ) and interfacial transferred charge ohmic resistance ( $R_{ct}$ ) which is independent of the electrode material and, therefore, comparable electrolyte conductivity and ohmic contact [33]. The semicircle at intermediate frequencies is linked to the charge-transfer resistance ( $R_{ct}$ ) in parallel with the capacitance of the double-layer, which can be given by a constant phase element (CPE) as a result of surface roughness and heterogeneity [34]. The inclined tail at low frequencies is the inventory of Warburg impedance (W), being the diffusion of  $\text{OH}^-$  ions across the electroactive nickel hydroxide layer [35].

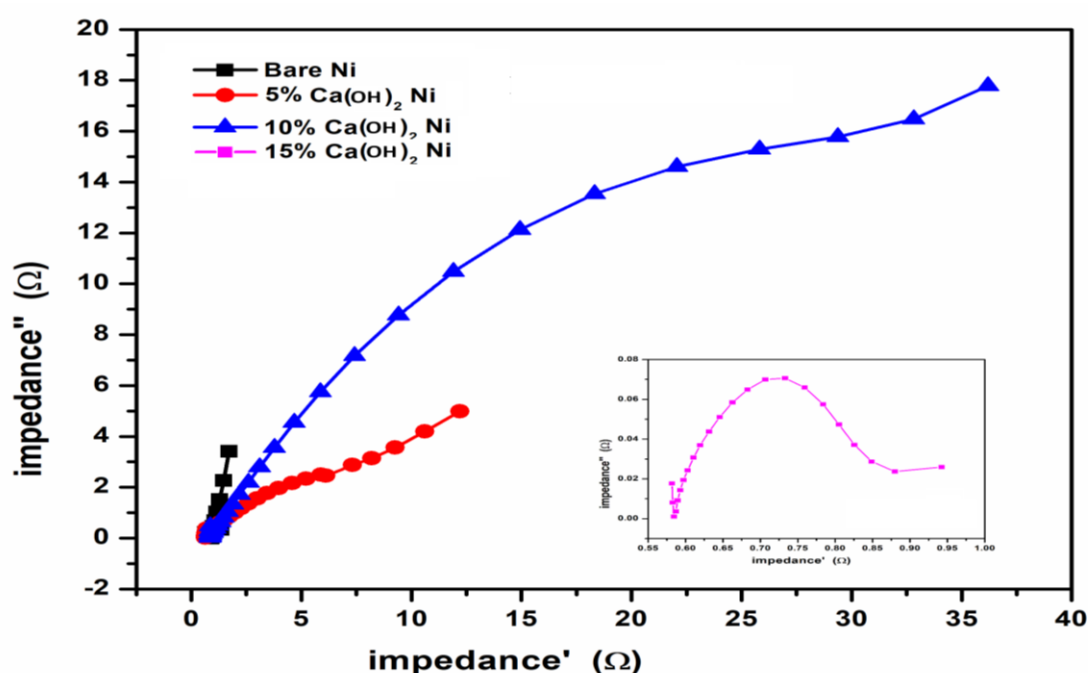


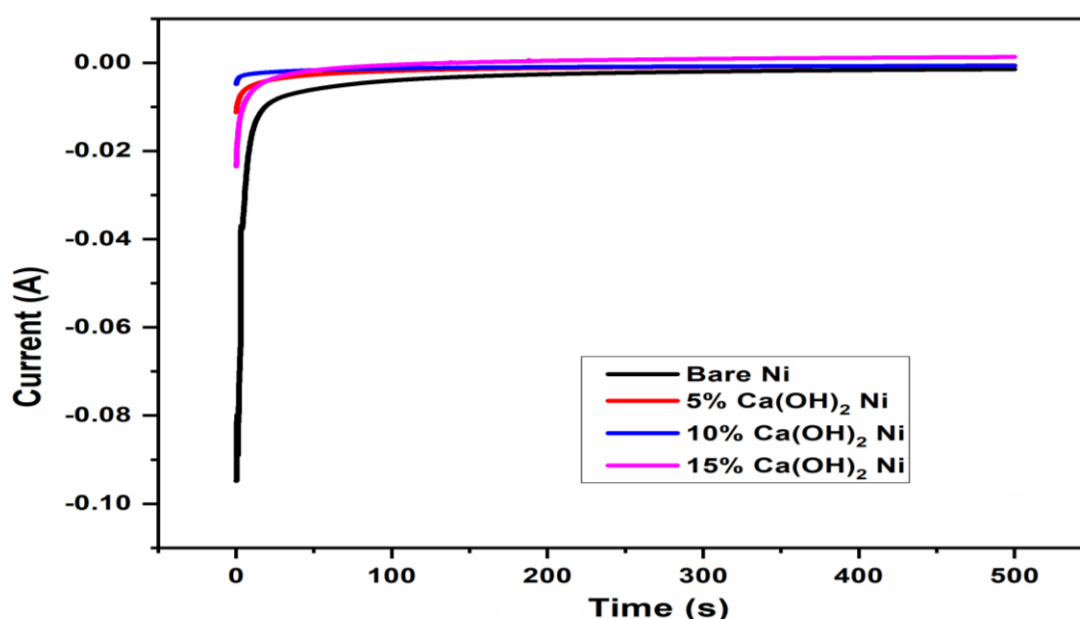
Figure 17. Nyquist plot for Ni electrode modified with 5%, 10% & 15%  $\text{Ca(OH)}_2$ .

The 5%  $\text{Ca(OH)}_2$  - modified Ni electrode has a comparatively wide semicircle diameter which indicates a higher  $R_{ct}$  and slower rate of charge transfer. Whilst the weak concentration of  $\text{Ca(OH)}_2$  increases local alkalinity, there is inadequate accessibility of OH at the nickel surface because coverage is limited at this concentration.

By comparison, the 10%  $\text{Ca(OH)}_2$  modified electrode exhibits a high degree of decreasing semicircle diameter, which corresponds to the lowest  $R_{ct}$  in the tested electrodes. This is a clear indication of an increased rate of interfacial charge transfer and an enhanced electrochemical reversibility. The maximized  $\text{Ca(OH)}_2$  coating increases the surface wettability, electrochemically active surface area, and efficient movement of  $\text{OH}^-$  ion, and thus stabilizes the  $\text{Ni(OH)}_2/\text{NiOOH}$  redox interface. A further

addition of  $\text{Ca(OH)}_2$  to 15% stabilizes this effect, but additional  $\text{Ca(OH)}_2$  addition appears to create a partially insulating layer that covers and inactivates active nickel and cannot conduct electricity. This fact indicates the presence of an ideal  $\text{Ca(OH)}_2$  loading after which the electrochemical performance begins to become poor.

Further information regarding the electrochemical profile and stability of the modified electrodes with regard to time is obtained by chronoamperometry (Figure 18). On the application of constant potential, all electrodes display a transient sharp response during the initials of current, which can be explained by the quickness of charging a double layer and a redox interaction of the surface. It is then succeeded by gradual current dissipation to a quasi-steady-state level of diffusion-controlled processes.



**Figure 18.** Chronoamperometric response of Ni electrode modified with 5%, 10% & 15%  $\text{Ca(OH)}_2$ .

The 5%  $\text{Ca(OH)}_2$  modified electrode displays a comparatively rapid decay in the current and a lesser steady-state current, which implies a limited electroactive surface area and poorest stability of the nickel hydroxide layer. The 10%  $\text{Ca(OH)}_2$  - modified electrode exhibits a higher initial current and a slower rate of decay but eventually constant maximum steady-state current density. Such a behavior is indicative of increased electrochemical stability, persistent redox activity and efficient diffusion of  $\text{OH}^-$  ions through an electrode structure. The enhanced current retention is associated with decreased polarization and enhanced resistance to passivation of the surface, in contrast, the altered electrode with 15%  $\text{Ca(OH)}_2$  exhibits faster current decay as well as a lower steady-state current, which is evidence of decelerated charge transfer and active site partial blockage by excess surface modifier.

The behavior at the present moment is controlled by the kinetics of the diffusion and can be described according to the Cottrell equation [18]:

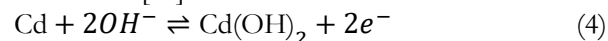
$$i(t) = \frac{nFAD^{1/2}C}{\pi^{1/2}t^{1/2}} \quad (3)$$

Electroactive surface area  $i(t)$  of a sample is the current at a given time  $t$ ,  $D$  is the diffusion coefficient of  $\text{OH}^-$  ions,  $A$  is the electroactive surface area, and  $C$  is the concentration of electroactive species. The increased and more stable current of the 10%  $\text{Ca(OH)}_2$  electrode means that there was a larger diffusion coefficient and higher surface area of the electrodes that was electroactive. The EIS and CA results have a strong correlation. Chronoamperometry Steady-state current and current retention are greatest with the electrode having the lowest  $R_{ct}$  (10%  $\text{Ca(OH)}_2$ ). pore diffusion resistance based on the region of low frequency impedance would be associated with the better stability of the long-term current. All these

complementary methods give evidence that optimized  $\text{Ca(OH)}_2$  modification has a significant positive impact on interfacial kinetics and mass-transport properties.

Cadmium plated on the stainless steel modified with  $\text{Ca(OH)}_2$  (Figure 19-25) and exposed to the voltammetry mode of cyclic voltammetry (CV) display the redox properties of negative electrode in alkaline Ni-Cd systems. Electrochemical process occurs under the influence of the reversible reaction

between metallic cadmium and cadmium hydroxide as follows: [36]:



The anodic peak is associated with the oxidation of Cd to  $\text{Cd(OH)}_2$  at the forward scan, and the cathodic one is associated with the reduction of  $\text{Cd(OH)}_2$  to metallic Cd in the reverse one in the CV profiles. The aspect of intensity and separation of these peaks gives an insight into the kinetics, reversibility and the surface activity of cadmium electrode.

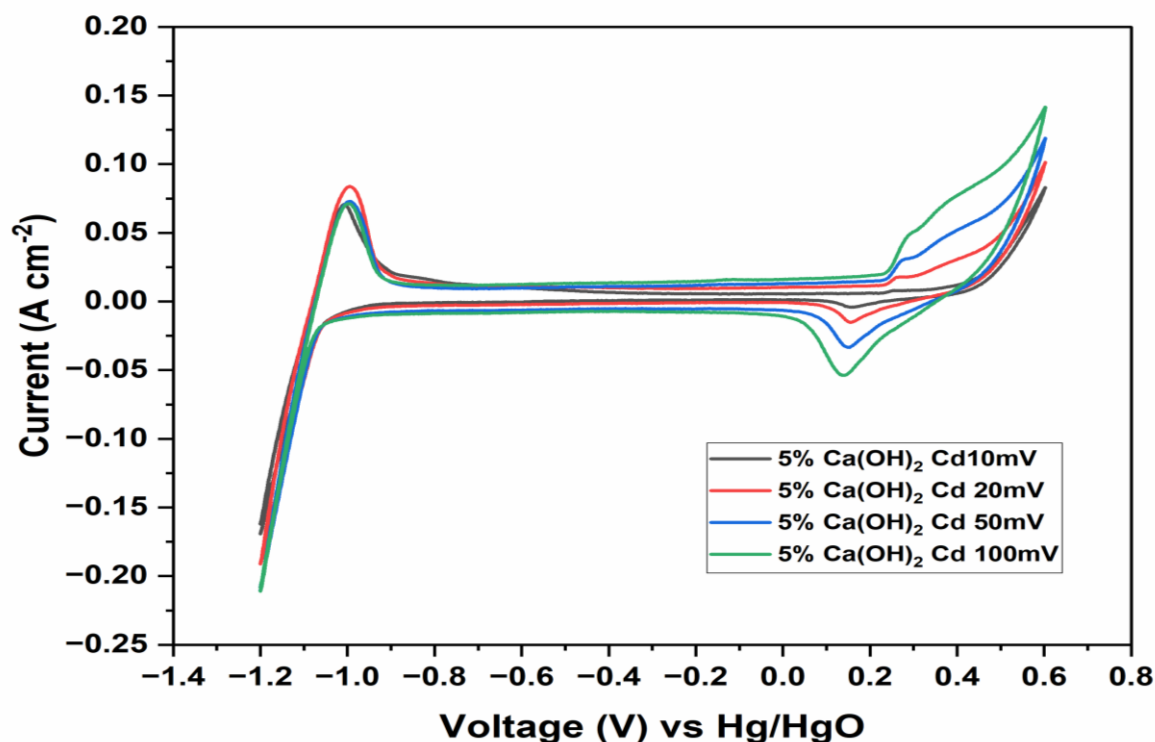


Figure 19. Cyclic voltammetric response of Cd electrode modified with 5%  $\text{Ca(OH)}_2$ .

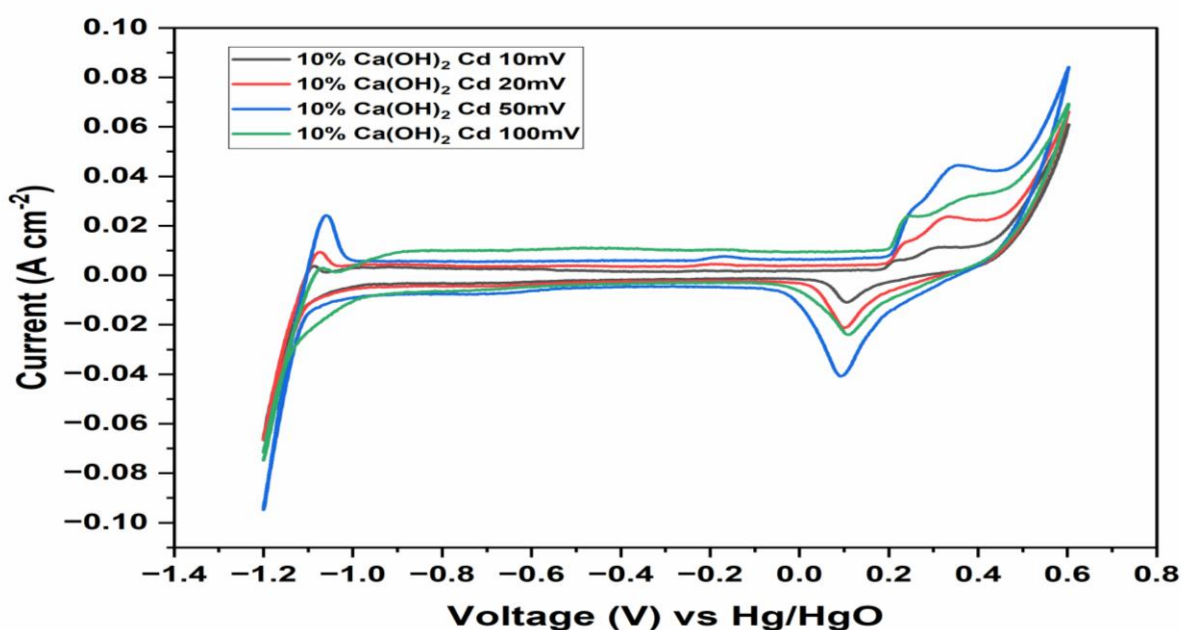


Figure 20. Cyclic voltammetric response of Cd electrode modified with 10%  $\text{Ca(OH)}_2$ .

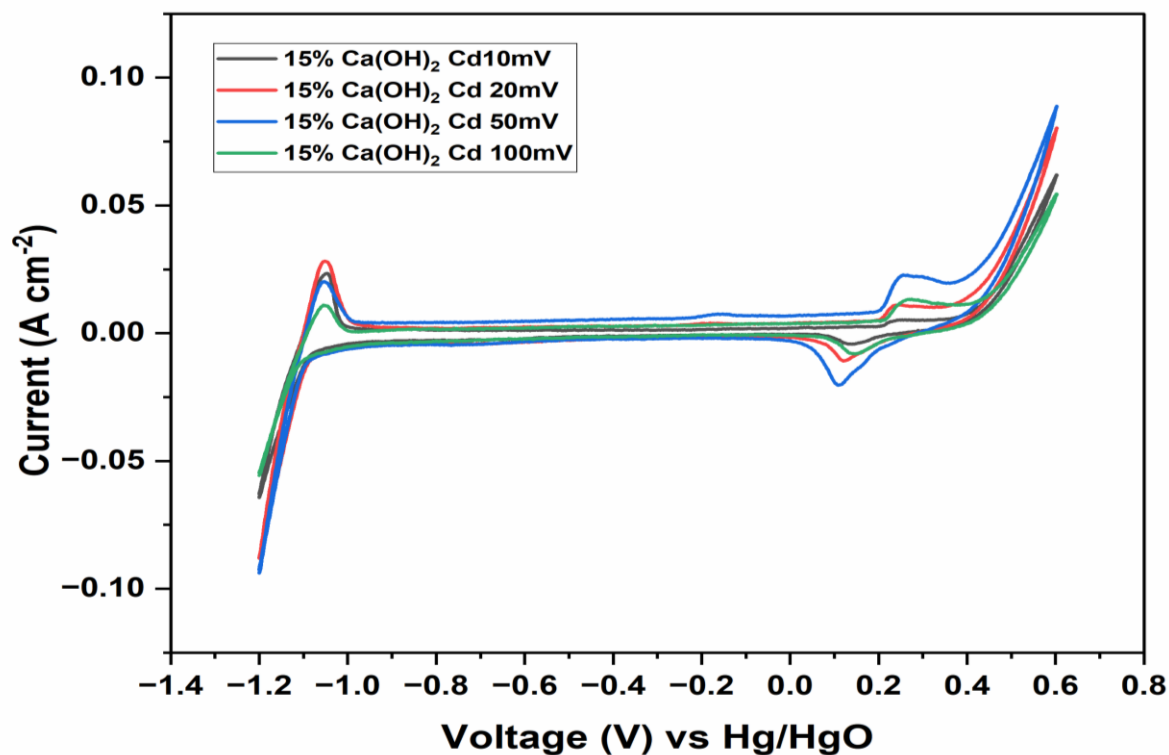


Figure 21. Cyclic voltammetric response of Cd electrode modified with 15%  $\text{Ca(OH)}_2$ .

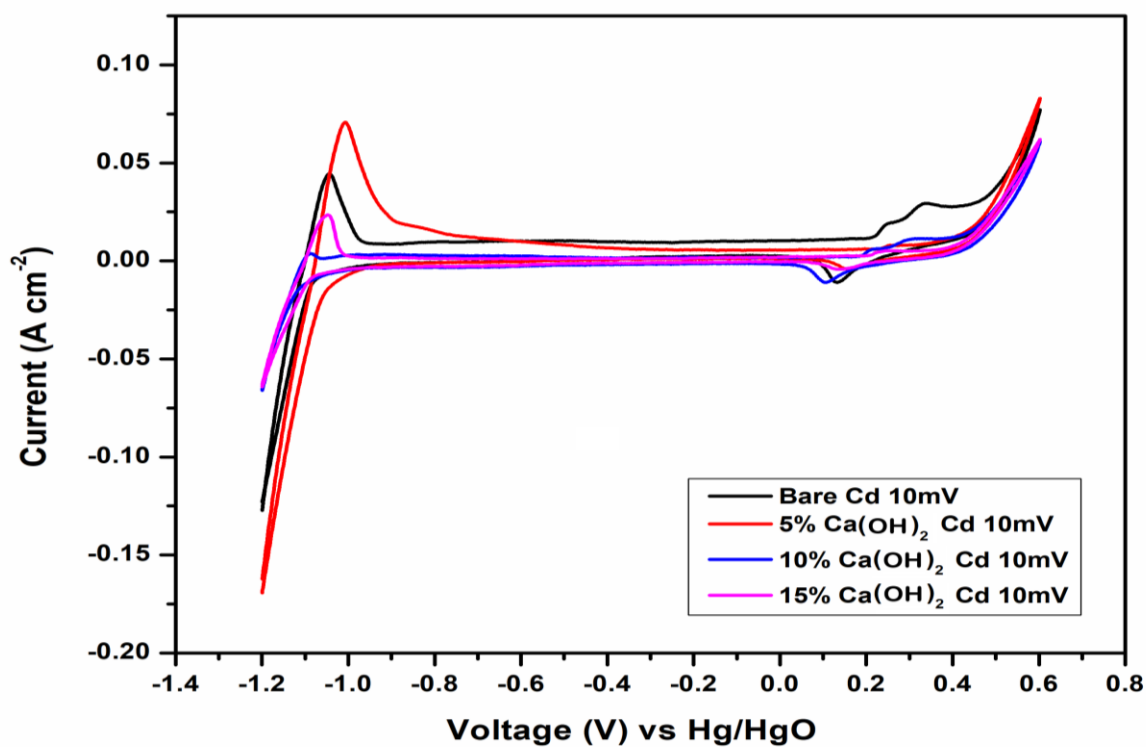


Figure 22. Cyclic voltammetric response of Cd electrode modified with 5%, 10% & 15%  $\text{Ca(OH)}_2$  at 10 mV/S scan rate.

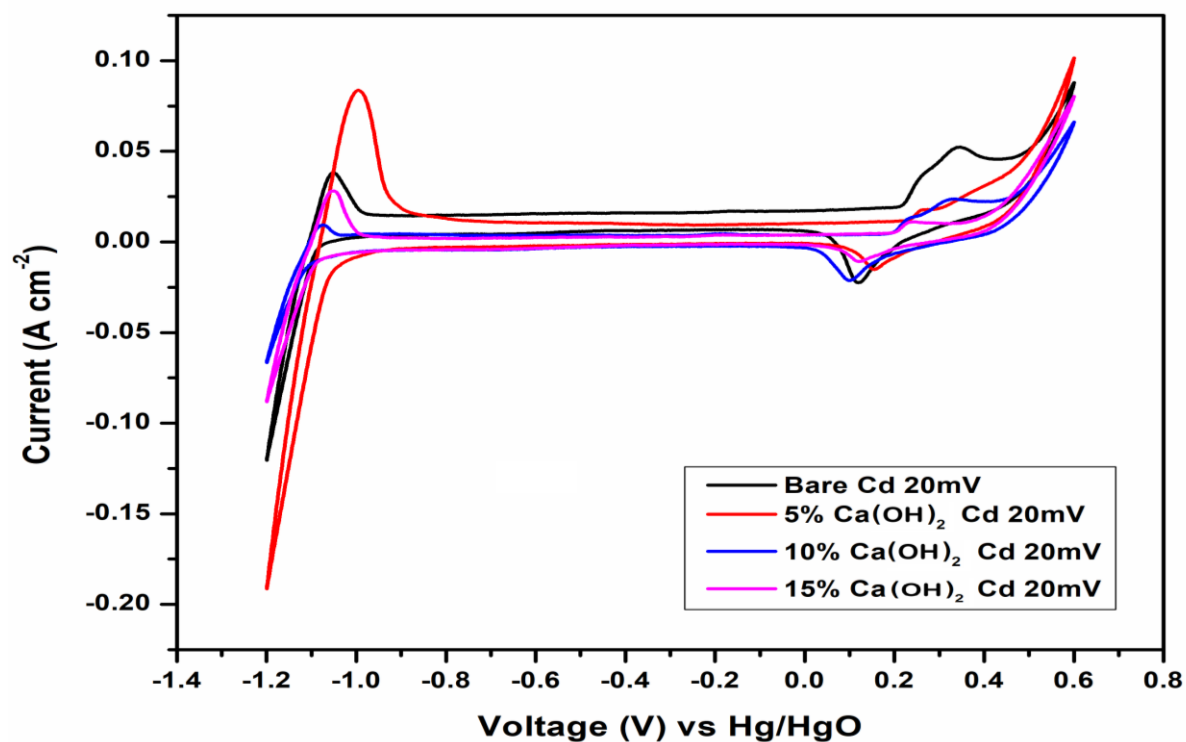


Figure 23. Cyclic voltammetric response of Cd electrode modified with 5%, 10% & 15%  $\text{Ca(OH)}_2$  at 20mV/S scan rate.

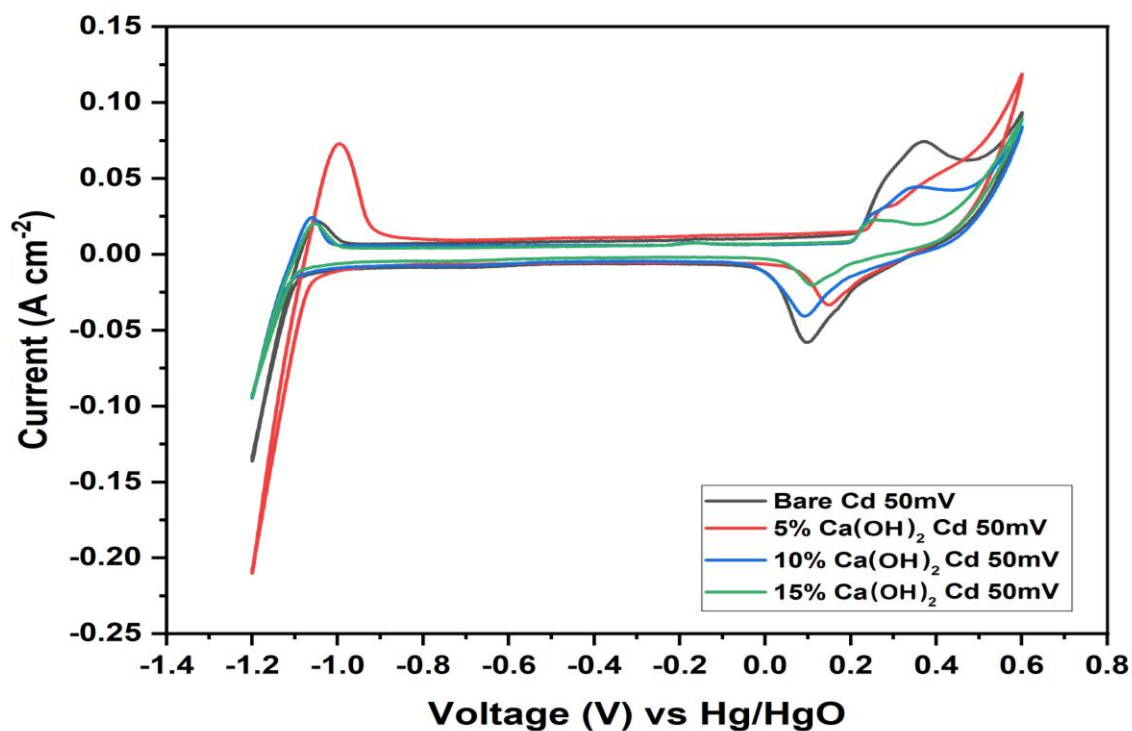
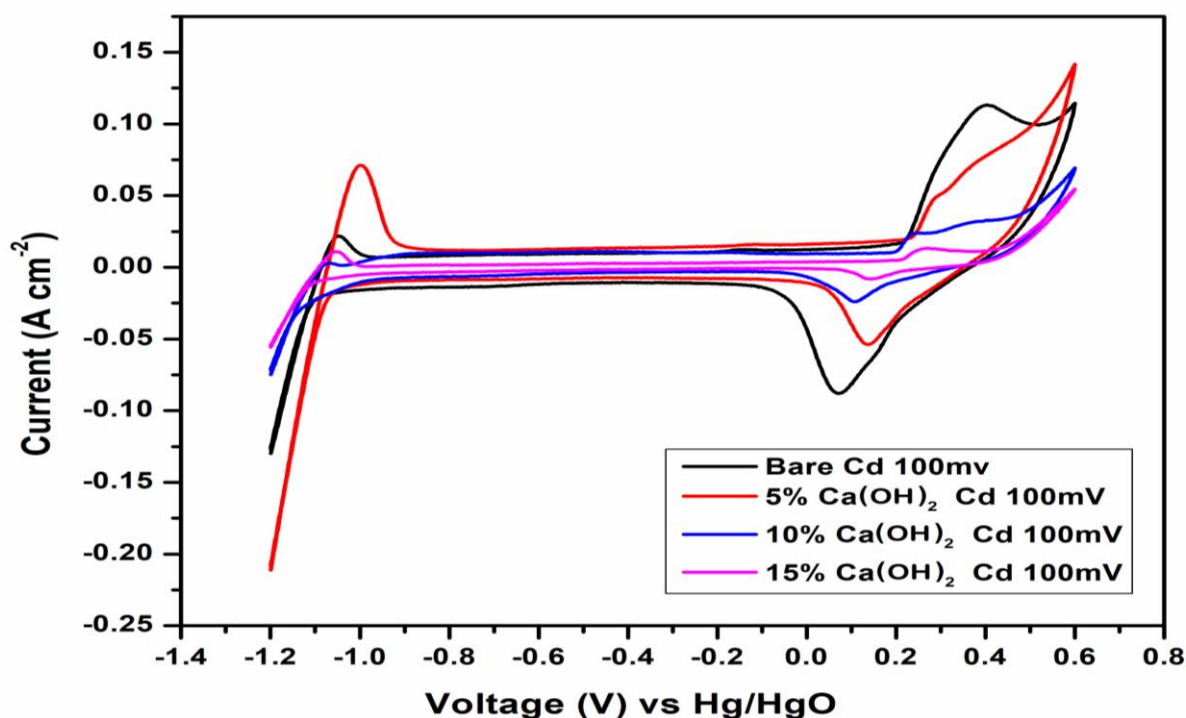


Figure 24. Cyclic voltammetric response of Cd electrode modified with 5%, 10% & 15%  $\text{Ca(OH)}_2$  at 50mV/S scan rate.





**Figure 25.** Cyclic voltammetric response of Cd electrode modified with 5%, 10% & 15%  $\text{Ca(OH)}_2$  at 100mV/S scan rate.

Figure 19 in the appendix has recorded specific anodic and cathodic peaks on the CV curve of the 5%  $\text{Ca(OH)}_2$ -modified Cd electrode, which validates the electrochemical accessibility of the  $\text{Cd/Cd(OH)}_2$  redox couple. Nonetheless, the comparatively low peak current density and the greater distance between the peaks imply moderate reversibility and greater polarization, which implies that either not all of the surface has been activated under this concentration of the  $\text{Ca(OH)}_2$  ions [37]. As the concentration of  $\text{Ca(OH)}_2$  increases to 10% (Figure 20), the electrochemical response is sharpened significantly. The anodic and cathodic peak currents grow and the optimal separation becomes smaller, which points out to enhanced reversibility and faster kinetics of charge-transfer. The optimized  $\text{Ca(OH)}_2$  coating increases local alkalinity, and the movement of  $\text{OH}^-$  ion, which offers even formation and reduction of  $\text{Cd(OH)}_2$ . This tendency is evidence of enhanced use of electroactive cadmium sites and lower interfacial resistance.

Conversely, the electrode that has been modified with 15%  $\text{Ca(OH)}_2$  (Figure 21) has a reduced peak current and considerable broadening of the peak. Surplus  $\text{Ca(OH)}_2$  would probably create a partially insulating surface layer that inactivates cadmium sites and thereby hinders the movement of electrons to cause suboptimal electrochemical activity. These findings show that there is an optimum  $\text{Ca(OH)}_2$  loading which maximizes the cadmium electrode activity.

Figures 22-25 demonstrate the influence of scan rate of the CV behavior of  $\text{Ca(OH)}_2$ -modified cadmium

electrodes. Figure 22 at a low scan rate of  $10 \text{ mV s}^{-1}$  shows redox peaks that are well and symmetric showing near equilibrium, and diffusion of  $\text{OH}^-$  ions. The greatest CV was encircling the 10% of  $\text{Ca(OH)}_2$ -modified electrode, which is the best in terms of charge storage abilities.

As scan rate increases to 20 and  $50 \text{ mV s}^{-1}$  (Figures 23 and 24), the amplitudes grow owing to higher reaction rates, and there are minor changes in the peaks, which are caused by polarization. Regardless of these changes, the 10%  $\text{Ca(OH)}_2$  electrode has greater values of current density, and more articulate redox characteristics, which attest to amplified reaction rates, as well as, to a better rate capacity.

When the scan rate was high and was  $100 \text{ mV s}^{-1}$  (Figure 25), it is possible to see the diffusion limitations in the case of the rapid polarization and high percentage changes: there is the peak broadening and between the 5%  $\text{Ca(OH)}_2$  and 15%  $\text{Ca(OH)}_2$ -modified electrodes, the peak separation becomes larger. However, the 10%  $\text{Ca(OH)}_2$ -modified electrode has a clear redox peak, which reflects its high electrochemical stability and high scan rate tolerance.

$\text{Ca(OH)}_2$  surface modification, in this case, is also important in the tuning of electrochemical kinetics of cadmium electrode. Moderate loadings of  $\text{Ca(OH)}_2$  increase local  $\text{OH}^-$  concentration and enhance electrolyte penetration and stabilize the  $\text{Cd/Cd(OH)}_2$  redox interface. All these effects minimize polarization losses and enhance homogeneous redox reactions. Nonetheless, a large amount of  $\text{Ca(OH)}_2$  presents interfacial resistance and restricts the

conductivity of electricity thus reducing the electrochemical activity.

The CV analysis shows that the alteration of  $\text{Ca(OH)}_2$  affects greatly the electrochemical characteristics of cadmium electrodes in Ni-Cd cells. The Cd electrode that was studied is the 10%  $\text{Ca(OH)}_2$  modified so that it carries the highest redox activity, better reversibility, as well as better rate performance. Such properties are the requirements to efficient charge acceptance and stable operation of Ni-Cd batteries implying that optimized  $\text{Ca(OH)}_2$  surface modification is a highly appealing approach to the improvement of negative electrode working.

Electrochemical impedance spectroscopy and chronoamperometry were used to compare the interfacial kinetics and stability of the cadmium electrode deposited on stainless steel/surface modified with  $\text{Ca(OH)}_2$ . The Nyquist plots (Figure 26) show intercept with high frequency, depressed

semicircle at middle frequencies and inclined tail with low frequencies, that are typical of cadmium electrode that is used with alkaline media. Such properties are related to the solution resistance ( $R_s$ ) and charge-transfer resistance ( $R_{ct}$ ) in parallel with a non-ideal double-layer capacitance and diffusion-controlled reactions related to the  $\text{OH}^-$  transport and the  $\text{Cd/Cd(OH)}_2$  phase change reaction.

Out of the electrodes examined, the 10%  $\text{Ca(OH)}_2$ -modified Cd electrode corresponds to the smallest semicircle diameter (or lowest  $R_{ct}$  and the highest kinetics of interfacial charge transfer). This enhancement is explained with optimized surface alkalinity, improved wettability and easier  $\text{OH}^-$  ion transport that, in turn, lead to more reversible  $\text{Cd/Cd(OH)}_2$  redox reaction:

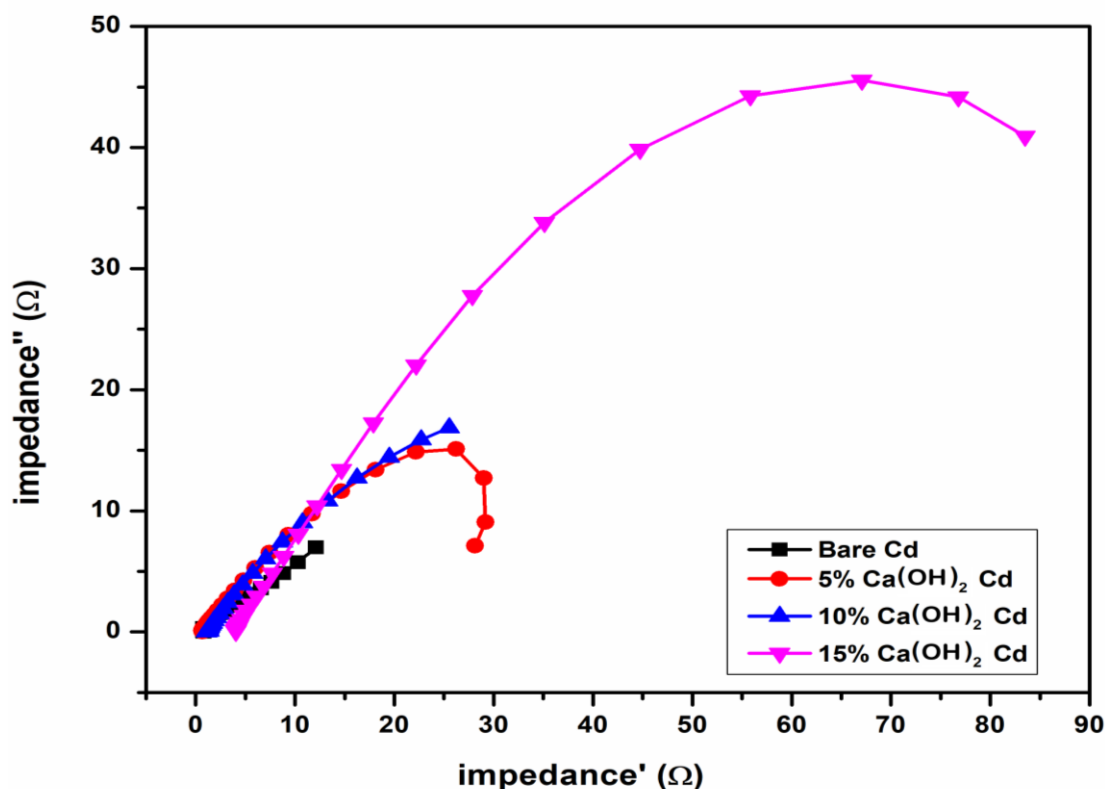
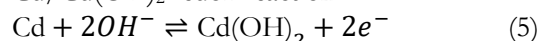


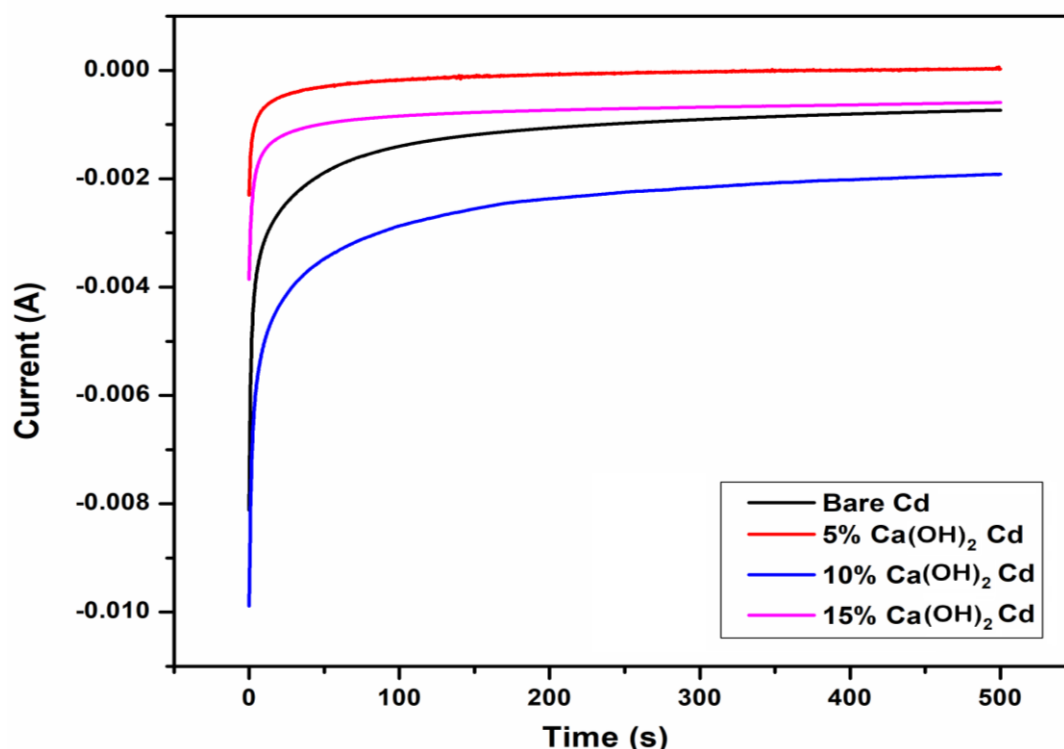
Figure 26. Nyquist plot for Cd electrode modified with 5%, 10% & 15%  $\text{Ca(OH)}_2$ .

Conversely, 5%  $\text{Ca(OH)}_2$  electrode will show increased  $R_{ct}$  because of inadequate surface activation whereas excessive modification at 15%  $\text{Ca(OH)}_2$  will result in interfacial resistance, probably because of partial wrap up of the electroactive cadmium sites by an isolating surface layer.

The impedance results are also supported by Chronoamperometry results (Figure 27). Each electrode showed an initial spike in the current

followed by the gradual decrease to an apparent steady-state burst, which is characteristic of faradaic processes that are diffusion-controlled. The 10%  $\text{Ca(OH)}_2$ -fixed electrode provides the high initial current and better output current retention, which corresponds to the increased electroactive surface area and decreased polarization. The example of the time-dependent behavior is Cottrell-type diffusion kinetics.





**Figure 27.** Chronoamperometric response of Cd electrode modified with 5%, 10% & 15%  $\text{Ca(OH)}_2$ .

The increased steady-state current obtained with the 10%  $\text{Ca(OH)}_2$  electrode indicates a greater diffusion coefficient effective and enhanced stability at the  $\text{Cd/Cd(OH)}_2$  interface. Either EIS or chronoamperometry show that the best electrochemical performance of cadmium electrodes is with 10%  $\text{Ca(OH)}_2$  surface modification, which has the lowest charge-transfer resistance, better diffusion properties and greater current stability [38]. The three properties play vital roles in the effective integration of  $\text{Ca(OH)}_2$  cells regarding charge acceptance and cycling stability under long-term conditions, as they prove the suitability of the controlled modification of the surface of negative electrodes using  $\text{Ca(OH)}_2$ .

Ni and Cd electrodes show a steady reliance on the surface modification of the  $\text{Ca(OH)}_2$ , and 10%  $\text{Ca(OH)}_2$  is found to be the best composition. This change reduces the charge-transfer resistance of both electrodes and increases the redox and the stability of the current compute. The synergistic enhancement of the positive and negative electrodes decreases losses in polarization and mediates the equal incorporation of charge discharges in the Ni-Cd cell.

On the whole, the combined CV, EIS, and CA studies also prove the effectiveness of controlled  $\text{Ca(OH)}_2$  surface modification in tuning electrochemical interfaces of purchased Ni electrode and Cd electrode resulting in an improved kinetics, improvement of diffusion characteristics and electrical stability. Such enhancements will be reflected in an increase in charge efficiency, rate capability and cycling stability of Ni-Cd batteries.

#### 4.CONCLUSION

In the present work, a combined electrochemical method was effectively designed to selectively recover cadmium and nickel of spent Ni - Cd cell electrolyte and to measure the performance of the recovered electrodes following the  $\text{Ca(OH)}_2$  surface modification. The systematic electrochemical studies with cyclic voltammetry (CV), linear sweep voltammetry (LSV), electrochemical impedance spectroscopy (EIS), and chronoamperometry (CA) accorded broad understanding on metal deposition behavior, interfacial charge, and interfacial stability of electrodes that made it possible to preferentially remove copper before Cd and Ni recovery. Application of complexing agents and ammonia media offered a shift in the possibilities of deposition and in the separation window of Cd and Ni validating the strength of the electro-separation policy even in intricate obtainable phases. of the electrode material discussed, stainless steel showed the most effective electrochemical properties, such as lower over potential, greater current response, lower charge transfer resistance and better reversibility, which makes it appropriate in the large-scale application.

$\text{Ca(OH)}_2$  surface modification of the recovered Ni and Cd electrodes had a significant effects on their electrochemical behaviours. It was found that 10%  $\text{Ca(OH)}_2$  modification maximizes redox activity, reversibility and rate capability in both  $\text{Ni(OH)}_2/\text{NiOOH}$  and  $\text{Cd/Cd(OH)}_2$  redox systems. It was found in EIS measurements, where the composition has the lowest charge-transfer

resistance, and in chronoamperometry, which shows enhanced current retention and passivation resistance to the surface. Reduced  $\text{Ca}(\text{OH})_2$  loading levels yielded incomplete surface recovery and excess modification augmented the interfacial resistance and diffusion constraints owing to partial blockage of electroactive locations, which caused reduced polarization loss in addition to a matched electrochemical performance at the anode and the cathode. Significantly, both Ni and Cd electrodes showed uniform and synergistic reactivity to the optimal  $\text{Ca}(\text{OH})_2$  concentration, which caused diminished polarization loss and diffusion constraints at the electrodes, respectively. These improvements directly apply to charge efficiency, rate performance and cycling stability of Nickel-Cadmium cells.

As a whole, this work creates a resource-effective and sustainable pathway of reclaiming, reusing, and improving performances of cadmium and nickel electrodes of used Ni-Cd batteries. Selective electro-separation coupled with an optimized surface modification is compatible with the principles of a circular economy, and it reduces the risks of environmental pollution of cadmium disposal, as well as it offers a potential option to prolong the lifecycle of the important battery material.

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